

F24-148-D-PromoHive

Project Team

Moiez Asif	21I-2483
Syed Muhammad Hassan Raza	21I-0465
Aasir Farrukh	21I-0375

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Supervised by

Dr. Faisal Cheema



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

Introduction

This document is intended as a mid-evaluation report of our Final Year Project (FYP), PromoHive, an AI-powered promotion engine. This FYP is being developed in collaboration with UpStart Commerce and Knowledge Discovery and Data Mining Lab (KDD) here at FAST Islamabad. This report is also intended to provide further insights into the scope of the project, the problem statement of the project, and the progress report. The readers of this report includes but are not limited to: developers, project managers, marketing teams, end users, testers, and technical writers, giving everyone a clear understanding of the system's current structure and future direction.

1.1 Problem Statement

The main problem that our project aims to solve is the reliance of marketers in e-commerce on multiple separate tools for customer segmentation, data analysis, and promotion management. Inefficient use of vast amounts of data is almost inevitable when these separate fragmented systems are used, and it leads to the promotions generated being too generic that do not resonate with the customers. Traditional e-commerce marketers also rely on static promotional calendars, which are reused and do not account for different customer segments. This inefficient use of data inevitably results in reduced customer engagement/satisfaction and sub-optimal sales. The problem we are addressing is a two-fold problem:

- **First fold of the problem:** The first fold refers to selecting the right discount type (qualifier) for the right customer. Different types of customers are appealed by different discount types based on their past behavior and purchase history. For example, get 50 percent off and buy one get one free is essentially the same thing, but a customer might think that he/she gets more value in 50 percent off as they might think that one free item in buy one get one free is not up to the standards. This

type of behavior is reflected in the customer's purchase history. A key challenge is identifying which discount type works best for each customer segment.

- **Second fold:** The second fold refers to determining the optimal discount percentage or the X and Y unknowns of the selected discount type in the first fold. This challenge includes deciding the right discount percentage that maximizes customer satisfaction/engagement while also maintaining profitability.

The following table provides a summary of 8 discount types that are commonly referred to in promotional strategies, as provided by the industry collaborator in this project: UpStart Commerce:

Discount Type	Description
Buy X Get Y Free	Offers customers a free product when they purchase a certain number of other products. (e.g. buy 2 get 1 free)
Fixed Price Discount	Sets a fixed price for a product or a set of products after meeting specific purchase requirements. (e.g. Get 100 dollars off on next purchase)
Percent Discount	Offers a discount based on a percentage of the product's price (e.g., Buy 1 for 10 percent off, Buy 2 for 20 percent off).
Each N-th Product Discount	Discounts every N-th product in the shopping cart, such as "20 percent off every 5th product."
Multiple Tiered Discount	Applies increasing discounts as the customer spends more or buys more (e.g., Spend 100 dollars, get 10 percent off; Spend 200 dollars, get 20 percent off).
Buy X Get Y with Fixed Discount	Offers a fixed discount on a product when a different product is purchased (e.g., Buy 1 shirt, get pants for 5 percent off).
Cart-Wide Discount	Offers a discount based on the total value of the shopping cart (e.g., Get 10 dollars off for every 100 dollars spent).
Product Set Discount	Provides a fixed discount or percentage off when a specific set of products is purchased together (e.g., Buy a couch, loveseat, and chair for \$999).

Table 1.1: Main Discount Types Provided by UpStart Commerce

1.2 Scope

Our project aims to develop a promotion engine that helps sellers create, manage, and automate promotional offers for different types of customer segments to achieve personalization. The main functionalities include a promotional calendar with customer segments, what type of promotion, and what percentage of discount to use for the promotion as its sub-features. Other features include automated emails and monitoring/reporting through dashboards.

Customer segmentation will analyze customer's purchase history/pattern and product preferences. Using machine learning, the system will suggest the most appropriate promotion type and the optimal discount value for the specific customer segment. These suggestions will be visible on the dashboards for the seller to approve/reject the promotion campaign. The dashboards would also show reports on past promotion performance that would be analyzed by the seller. Furthermore, the engine would allow sellers to send automated personalized emails to their customers for promotions.

The boundary of the engine is that it would not be using any external or competitor data. It will focus only on the company's data. Similarly, all the recommendations would be based on past data and not on real-time. Also, the engine would not be considering individual customer preferences at a level beyond the predefined customer segments. Additionally, our engine will be based upon MACH principles - Microservices, API-first, Cloud-native, Headless and we are focusing on producing a Headless product. By headless, we mean that there will be a minimal UI and our main operations will be performed via APIs. Moreover, the engine does not include dynamically adjusting promotions in real-time, and the dashboard would not have live reporting. The algorithm focus will be on machine learning algorithms like Regression, Clustering, Decision Trees, Random Forests, etc.

1.3 Literature Review

1.3.1 Paper 1 (2018)

Author: Hong Xie, Richard T. B. Ma (National University of Singapore), John C. S. Lui (Chinese University of Hong Kong)

Title: Enhancing Reputation via Price Discounts in E-Commerce Systems: A Data-Driven Approach

Objective/Purpose: To create a framework for discount rate optimizing to balance profits

and maintain seller's reputation.

Methodology: To determine the best discount rates, authors developed a stochastic bandits framework. They used eBay data to understand buyer behaviors and preferences towards discounts.

Key Findings/Results:

- Sellers reduced their time by 40% while losing 60% of short-term profit.
- Long-term profit increased by at least 20% with the discount strategy.
- The algorithm balanced losses with gains proving beneficial for new sellers who are looking to increase their reputation.

Gaps/Limitations: The study focused on new sellers and may not be as useful to established businesses.

Relevance to Our Research: This research paper is talking about discount optimization which can be useful for our project.

1.3.2 Paper 2 (2022)

Author: Soravit Wetprasit (University of Glasgow), Qi Cao (University of Glasgow), Chee Kiat Seow (University of Glasgow)

Title: Recommender System for Coupon Discount of E-commerce Applications

Objective/Purpose: To develop a hybrid recommender system for coupon discount recommendations in e-commerce applications.

Methodology: To match coupons to client preferences, the authors used matrix factorization (Alternating Least Squares) and they also used association rule mining (Apriori method). They used customer-item interactions and evaluated using AUC and Precision at K metrics to assess the model.

Key Findings/Results:

- The model showed a 20% improvement in precision compared to the normal existing models.
- It achieved an AUC score of 0.912 which means that it is performing better than feedback models.
- The used model ranked the most relevant coupons at the top, with precision of 0.7 with K set to 5.

Gaps/Limitations: They did not go deep into understanding the personal behavior of customers.

Relevance to Our Research: As it combines customer demographics, transaction history, and implicit feedback. This may help us in doing customer segmentations.

1.3.3 Paper 3 (2019)

Author and Year: Mei and Li (2022)

Title: Research on E-commerce Coupon User Behavior Prediction Technology Based on Decision Tree Algorithm

Objective/Purpose: To develop a predictive model for coupon redemption behavior in e-commerce to enhance targeted coupon distribution.

Methodology: The authors used XGBoost (eXtreme Gradient Boosting) to create a prediction model which is based on decision trees. To optimize the model, they applied feature engineering and parameter tuning on Alibaba's O2O e-commerce data.

Key Findings/Results:

- With an accuracy of 0.9206 and an AUC of 0.8223, the XGBoost model performed well.
- Coupon utilization within a 15-day window was anticipated well by the model.

Gaps/Limitations: They only focused on a small time span of 15 days which may not capture long term customer behavior.

Relevance to Our Research: This research provides valuable insights into predicting coupon usage behavior, which is important for our AI-based promotion system. We would understand better how the prediction model can determine the customer segmentation for optimal discounts.

1.3.4 Paper 4 (2024)

Author: Anitha Julian, Challa Maheswar, Ramyadevi R, Murali Dhar M S, Selvi S

Title: Analysis and Estimation of Discounts on Product Prices in E-Commerce Sites

Objective/Purpose: To develop a framework for analyzing and estimating optimal discounts for products in e-commerce platforms.

Methodology: The authors used the exploratory data analysis (EDA) to identify the existing patterns and the relations between product sales and discount. They used the available

historical transaction data from e-commerce platforms to develop predictive models for discount optimization.

Key Findings/Results:

- Identified important relations between discount levels and sales volume.
- Certain product categories consistently received larger discounts which increased sales.
- The predictive model demonstrated high accuracy in estimating sales based on discount levels.

Gaps/Limitations: The study did not make use of external factors like trends.

Relevance to Our Research: This research provides a lot of explanation between discount and sales. Using this information we can benefit by determining the X and Y values in our discount type calculation.

1.3.5 Paper 5 (2022)

Author: Gaurav Jangra, Monika Jangra

Title: Role of Artificial Intelligence in Online Shopping and its Impact on Consumer Purchasing Behaviour and Decision

Objective/Purpose: Understanding the use of Artificial Intelligence on customer purchasing behavior.

Methodology:

- Descriptive research design
- 200 respondents were given a structured questionnaire across a few Haryana districts.
- Statistical analysis: Descriptive statistics, Chi-Square, ANOVA, and regression analysis

Key Findings/Results:

- AI explains 65% of the variance in customer purchasing decisions with R^2 value of 0.650
- Strong impact of AI on purchasing behavior could be seen with $p\text{-value} < 0.05$

- ANOVA analysis showed the importance of the regression model with F-value = 367.26, p-value < 0.05

Gaps/Limitations:

- Only few districts from Haryana were selected so there is an issue of limited scope
- There might be bias due to the conventional sampling method

Relevance to Our Research: This study provides useful insights into the relationship between AI and consumer behavior in online shopping. This is important for our AI Promotion system. This would benefit us in doing customer segmentations when we can understand what type of factors impact customers by how much amount.

1.4 Modules

1.4.1 Module 1

This module will create promotions for different types of customer segments. A promotion calendar would be generated, which would show what promotion type with what discount value to be applied to different customer segments.

1. AI Powered Promotion Calendar
2. Customer Segmentation
3. Customizable Discounting

1.4.2 Module 2

This module will use the output from the previous module and allow the company to send emails to the different customer segments. Moreover, it will allow the company to analyze and optimize its promotional strategies.

1. Automated Emails
2. Dashboards

1.4.3 Module 3

This module will handle the deployment of our promotion engine and allow seamless integration with the e-commerce platforms. The integration will make sure that the engine is flexible (handles different types of data) and scalable (handles different amounts of data).

1. Deployment
2. System Integration

1.4.4 Module 4

The last module focuses on the final stage of the project. Rigorous testing will be done to ensure all functionalities are working as expected. Based on the feedback received, fine-tuning will be applied to improve performance.

1. Offline A/B Testing
2. Fine-Tuning

1.5 User Classes and Characteristics

User Class	Description
Marketers and Retailers	These users will be responsible for reviewing and approving promotions. They will also have access to a dashboard for reviewing a promotion's effectiveness via visuals.
Development Team	This group will handle the development, deployment, and maintenance of the promotion engine. They will also ensure that the promotion engine is successfully integrated with different e-commerce platforms.
Customers	The customers include the end users who will receive personalized promotions generated by our AI-powered promotion engine, aiming to enhance their shopping experience.
System Administrators	This group will be responsible for the overall management of the system.

Table 1.2: User Classes and Characteristics

Chapter 2

Project Requirements

2.1 Use-case/Event Response Table/Storyboarding

The use cases of our AI-based promotion engine are listed below:

1. **Generate Promotional Calendar:** The AI-based engine will automatically create a promotion calendar for marketers to view. This calendar will generate promotions for different customer segments, taking into account the segment's purchase history and behavior.
2. **Customer Segmentation:** Our AI promotion engine will analyze customer data, purchase history, and cart data, grouping customers into segments for targeted/personalized promotions.
3. **Dynamic Discount Type Classification:** The engine will dynamically allocate a discount type to each customer segment, relying on purchase history, behavior, preferences, and purchasing patterns.
4. **Dynamic Discount Value Calculation:** Continuing from the previous use case, the engine will calculate the optimized discount value for the selected discount type, balancing customer satisfaction and business profitability (profit margin).
5. **Automated Promotion Execution:** The proposed system will automate the entire promotion process, from segmentation to rolling out the promotion, without human intervention. This includes data analysis, customer segmentation, selecting the promotion type, and optimizing the discount value.
6. **Email Campaigns:** Automated and personalized emails will be sent based on the approved promotion calendar, boosting customer engagement and increasing business profit.

7. **Monitor Promotion Performance:** A dashboard will be provided for marketers to monitor promotion performance, enabling data-driven decisions.
8. **Offline A/B Testing:** Marketers can test different scenarios on existing datasets and compare outcomes without running live experiments.
9. **Marketer Approve or Reject Promotions:** Admins/marketers will have the authority to approve or reject promotions generated by the engine before they are rolled out.

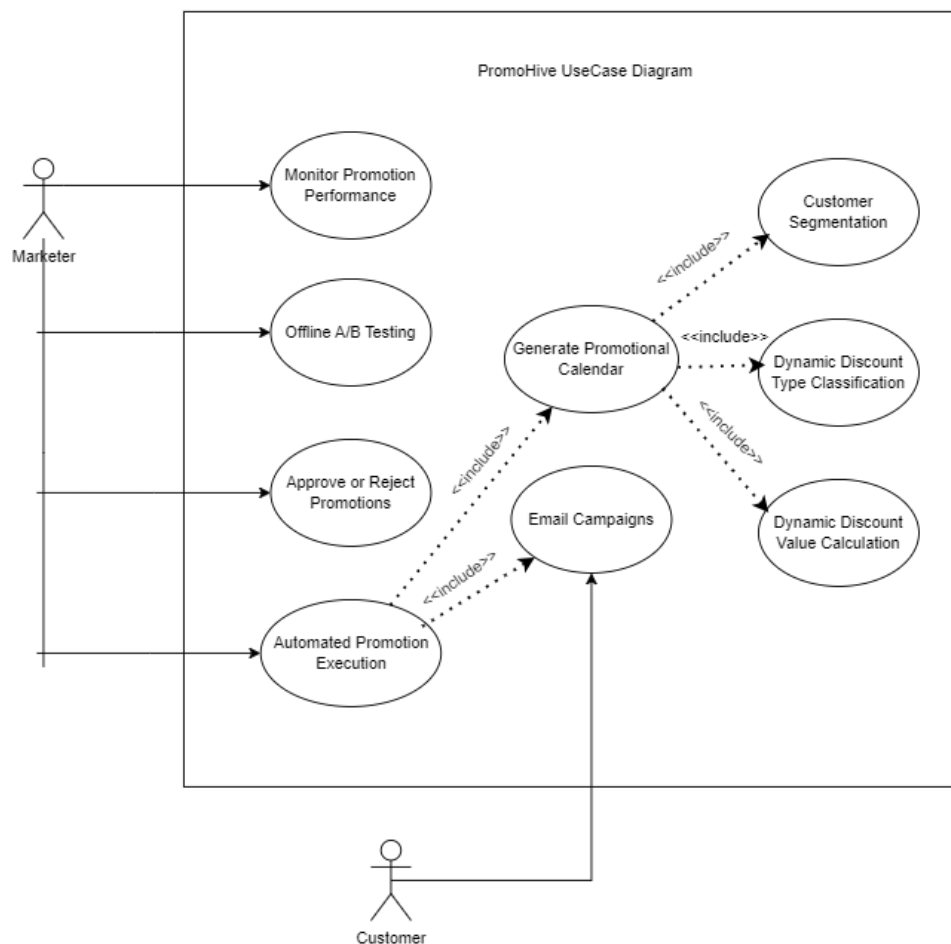


Figure 2.1: Use-case Diagram.

2.2 Detailed Use Cases

Building upon the use cases listed in the previous section, this section contains detailed descriptions of each use case.

2.2.1 Use Case 1

Use Case Name	Generate Promotional Calendar
Scope	AI Promotion Engine
Level	User goal
Primary Actor	Marketer
Stakeholders and Interests	• Marketer: Expects a visually appealing and easy to understand promotional calendar with optimized promotions for different customer segments to increase engagement and profitability. • Business: Businesses want personalized promotions to increase their revenue, sales, profit margin, and average sales per day. • Customers: Customers want personalized promotions to connect with businesses on a personal level, and those promotions have to take into account the history, purchase pattern, and preferences.
Preconditions	• The database, or PostgreSQL in our case, must have the whole customer data. This data includes purchase history, cart behavior, and preferences. • The marketer must be logged into the system. • Segmentation, regression, and classification models must be in place and active.

Success Guarantee (Postcondition)	The system generates a detailed, visually appealing, and easy to understand promotional calendar and that calendar should show optimized promotions for each customer segment.	
Main Success Scenario	Actor Action	System Responsibility
	The marketer logs into the system.	The system authenticates the marketer.
	The marketer navigates to the "Generate Promotional Calendar" section.	The system displays the "Generate Promotional Calendar".
	The AI-based promotion engine must be active and rolling and it will query the customer data, when the marketer clicks on "Generate button".	The AI-based promotion engine queries the customer data.
	The system processes the data.	The queried customer data will then be analyzed and passed through segmentation, regression, and classification to obtain results.
	Using segmentation, regression, and classification techniques, the engine will then populate the promotional calendar.	The calendar is populated with promotions, discount types, discount values, and timestamps for each customer segment.
	The marketer sees and reviews the promotion calendar.	The system displays the promotional calendar in a visually appealing and understandable format for review.

Extensions (Alternate Flows)	• E1: If certain data points from customer data are unavailable for certain segments: – The system will use external data resources or use predefined business rules to create generalized promotions. – The marketer will be notified that the data used for certain segments is not fully personalized.
Special Requirements	• The calendar must be visually appealing and easy to understand, and the user(marketer) must be able to filter the calendar based on customer segment or time period. • The promotion engine should keep in mind the profit margins whether the promotions are hurting those margins a lot and whether the promotions also capture the customer's interest.
Technology and Data Variations	• The system should be able to handle data of various sizes and formats from numerous e-commerce platforms. It should also account for different date formats for sales data.

2.2.2 Use Case 2

Use Case Name	Customer Segmentation
Scope	AI Promotion Engine
Level	Sub-function
Primary Actor	AI-powered promotion engine
Stakeholders and Interests	• Marketer: Marketers need multiple yet precise customer segments that understand and capture the customers' behaviour accurately. • Customers: Customers need themselves to be placed into segments that are related to their behaviour. • Business: Businesses want their promotions to be personalized and targeted so that they can connect with the customers on personal terms and increase their profitability.
Preconditions	• The database or PostgreSQL in our case must have the whole customer data. This data includes purchase history, cart behaviour, and preferences. • Segmentation model must be in place and running.
Success Guarantee (Postcondition)	• The promotion engine must correctly and accurately segment each customer into groups and that segmentation is done based on demographics, purchasing pattern/history and customer behaviour.

Main Success Scenario	Actor Action	System Responsibility
	The AI-powered promotion engine receives a specific customer's data.	The system queries the customer data from the database.
	The engine then queries the customer data from the database and uses that info i.e. purchasing history, cart history and behaviour and analyzes that data.	The system processes the data using segmentation techniques.
	using the trend present in data, the AI segmentation model will classify each person into different groups.	The system creates customer segments based on demographics, purchasing patterns, and behavior.
	The end user(marketer/business) can see the segments of their customer base.	The system displays the segmented customer groups.
Extensions (Alternate Flows)	• E1: If certain data points from customer data are unavailable e.g marital status etc.: – Then the system will make use of only the data points that are available e.g. demographics and then group the customers based on that. – The user will be notified about the missing data points.	

Special Requirements	• It is crucial that the segmentation model produces an output that is fast, scalable, and can handle all sizes and formats of data. • Segments can be updated in real time as soon as new data becomes available.
Technology and Data Variations	• The data points will vary in size, format, and availability from one e-commerce platform to another.

2.2.3 Use Case 3

Use Case Name	Dynamic Discount Type Classification
Scope	AI Promotion Engine
Level	Sub-function
Primary Actor	AI-powered promotion engine
Stakeholders and Interests	• Marketer: Marketers want help identifying the right type of discount for the right kind of customer segment to ensure maximum customer engagement. • Customers: Customers want their shopping habits to be identified and learned to give them personalized promotions according to it. • Business: Businesses want their profitability to be maximized but they also want to give market competitive discounts without hurting their profit margins.
Preconditions	• Customer segmentation must be done keeping in mind the customer purchase pattern, purchase history, and behavior. • Customer segmentation must be completed and the results must stored in a database. • The system's discount rules and business goals should be defined.
Success Guarantee (Postcondition)	• The customer segment must be assigned the right and most resonating discount type e.g. Buy one get one, tiered discount according to the customer's data.

Main Success Scenario	Actor Action	System Responsibility
	The AI-powered promotion engine accesses the segmentation data stored in the database.	The system queries the customer data, including purchasing history, cart behavior, preferences, and customer behavior.
	The AI promotion engine selects the best suited discount type for each segment based on previous discount type data and/or predefined business rules.	The system analyzes the customer data.
	The end user(marketer/business) can see the breakdown of discount types allocated to each customer segment.	The system displays the assigned discount types for each segment.
Extensions (Alternate Flows)	• E1: If certain data points from customer data are unavailable e.g marital status etc. or customer data is insufficient then: – Then the system will apply default business rules to assign a discount type. – The user will be notified about the missing data points and/or customer data being insufficient.	

Special Requirements	• It is extremely important that the discount type selected does not hurt the business's profit margins or start to incur loss. • Such a discount type has to be selected so that a balance between customer satisfaction and profitability is reached.
Technology and Data Variations	• The assigned discount type will have variations for example percentage discount is allocated for new visitors/customers and a tiered discount type is selected for old customers.

2.2.4 Use Case 4

Use Case Name	Dynamic Discount Value Calculation
Scope	AI Promotion Engine
Level	Sub-function
Primary Actor	AI-powered promotion engine
Stakeholders and Interests	• Marketer: Marketers want to identify and achieve the right value of the discount to maintain an optimum balance between business profitability and customer satisfaction. • Customers: Customers want the exact amount of discount to complete the purchase • Business: Businesses want their profitability to be maximized but also they need to be ensured that the profit margin stays high and/or in the acceptable range.
Preconditions	• Customer segmentation must be done keeping in mind the customer purchase pattern, purchase history, and behavior. • Customer segmentation must be completed and the results must stored in a database. • In addition to customer segmentation, the allocated discount type to each customer segment must also be stored in the database which can be queried to access it. • The customer data, business goals, and systems' discount rules should be defined.
Success Guarantee (Postcondition)	• After considering the optimum balance between customer satisfaction and business profit margins, an optimal discount value is assigned to the discount type template selected.

Main Success Scenario	Actor Action	System Responsibility
	The AI-powered promotion engine accesses the segmentation data stored in the database. It further accesses the discount type assigned to each customer segment.	The system queries customer data, including purchasing history, cart behavior, preferences, and behavior.
	The engine then calculates the right discount value for the given discount type by querying the customer data from the database and using that info i.e. purchasing history, cart history, preferences, and behavior, and analyzing that data.	The system takes into account the business's profit goals during the calculation.
	The end user(marketer/business) can now see the discount value assigned to each discount type of each customer segment and can now approve/reject it.	The system displays the discount values for review and approval/rejection.

Extensions (Alternate Flows)	• E1: The selected discount value by the model/engine is not satisfying the profitability goals and instead exceeding the discount threshold: – Then the engine will make the discount value reduced to ensure business's profitability. – The user will be notified about the adjustment in discount value made.
Special Requirements	• The calculated discount value must ensure business's profitability and customer satisfaction, also it should take into consideration the demand elasticity and profit margins.
Technology and Data Variations	• The discount value will be variable by taking into account the demand for the product.

2.2.5 Use Case 5

Use Case Name	Automated Promotion Execution
Scope	AI Promotion Engine
Level	User Goal
Primary Actor	AI-powered promotion engine
Stakeholders and Interests	• Marketer: Marketers want the entire promotion process to be automated from segmentation to execution, to eliminate the need for manual and mundane tasks which are done by humans and prone to errors. • Customers: Customers want the promotions to reach to them in a timely and accurate manner. • Business: Businesses want their profitability to be maximized without the need for manual labor.
Preconditions	• The promotions including the segment, discount type, and discount value must be approved and ready to roll out. • All of the data related to promotions i.e. segmentation, discount types, and values must be configured.
Success Guarantee (Postcondition)	• The promotions that are approved and are assigned by segmentation, discount type assignment, and discount value assignment are automatically rolled out with minimal to no human intervention.

Main Success Scenario	Actor Action	System Responsibility
	The AI-powered promotion engine retrieves the promotions generated.	The system identifies the promotions assigned to each customer segment along with their discount values.
	The promotions are rolled out and each customer receives a promotion that is specifically tailored for them.	The system rolls out the promotions.
	The marketer/business can track and review the promotion execution.	The system records all the steps it takes so that it can analyze its performance afterwards and those records can be used to enhance the engine further in the future.
Extensions (Alternate Flows)	• E1: If some data inconsistency is found and promotions fail to roll out: – The error is entered into the system log. – The user will be notified about the error.	
Special Requirements	• The automation must be seamless and scalable, meaning it should roll out promotions without any issues and can handle large scale as well as small scale promotion data.	
Technology and Data Variations	• The promotion should be delivered automatically through various channels.	

2.2.6 Use Case 6

Use Case Name	Email Campaigns
Scope	AI Promotion Engine
Level	User Goal
Primary Actor	AI-powered promotion engine
Stakeholders and Interests	• Marketer: Marketers want the promotional calendar that has been generated to be aligned with the automated email promotion campaigns. • Customers: Customers want the promotions they receive via emails to be personalized and resonate with them. • Business: Businesses want their profitability to be maximized with the email promotions.
Preconditions	• The promotions including the approved segment, discount type, and discount value must be approved and ready to roll out. • The email template should be ready and saved into the database.
Success Guarantee (Postcondition)	• The promotions that are approved are sent to the right customers at the right time via emails.

Main Success Scenario	Actor Action	System Responsibility
	Using the approved promotions, our AI engine will generate personalized promotional emails.	The system retrieves customer data and personalizes the email content with the relevant promotion details.
	Promotions are rolled out via email to the right customer segments.	The AI system personalizes the email with customer information and promotion specifics.
	The marketer/business can monitor the email campaign's progress.	The system records all the steps it takes so that it can analyze its performance afterwards and those records can be used to enhance the engine further in the future.
Extensions (Alternate Flows)	• E1: If the emails fails to be sent to the customer: – The system will try again to send the email and log the failure of sending the email. – The user will be notified about the failed delivery.	
Special Requirements	• The email promotion rolling out must be adapted for receiving the email on various platforms e.g. mobile, desktop, etc. and it should also comply with the GDPR regulations for customer data handling.	
Technology and Data Variations	• Email formats and way of language must be adapted according to different regions and preferences of the customer segment.	

2.2.7 Use Case 7

Use Case Name	Monitor Promotion Performance
Scope	AI Promotion Engine
Level	User Goal
Primary Actor	Marketer
Stakeholders and Interests	• Marketer: Marketers want to receive insights based on the promotion's recent performance after roll out. • Business: Businesses want monitoring of the promotions to make informed decisions which will lead to increased profitability.
Preconditions	• The promotions must have been active once in the past or must be running currently.
Success Guarantee (Postcondition)	• The system receives the insights from the promotion's performance and presents those insights in a visual manner for the marketer to understand and comprehend.

Main Success Scenario	Actor Action	System Responsibility
	The user(marketer) opens the analytics dashboard.	
	The marketer views the dashboard.	The system queries the promotion data from the database. The system then generates a dashboard for the user to see, the dashboard must have metrics such as click-through rates, conversion rates, and sales generated.
	The end-user(marketer) can see and review the visual representation of data and make informed decisions based on it.	The system provides a visual representation of performance data (e.g., graphs, charts) for easy comprehension.
	The marketer makes adjustments based on the insights.	The system also logs and maintains the changes made by the marketer for further fine-tuning the model in the future.
Extensions (Alternate Flows)	• E1: The promotion data is incomplete or the promotion is not found: – The system will show a warning message that the promotion data is insufficient or not found.	
Special Requirements	• The dashboard must have a user-friendly interface and must be interactive. • The dashboard must provide valuable insights. • The dashboard must contain visual graphs like pie charts and/or bar graphs for easier understanding and comprehension of the insights.	

Technology and Data Variations	• The medium of delivery e.g. email promotion or in-app promotion will determine the variation in data that has to be displayed.
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2.2.8 Use Case 8

Use Case Name	Offline A/B Testing
Scope	AI Promotion Engine
Level	User Goal
Primary Actor	Marketer
Stakeholders and Interests	• Marketer: The marketer does not want to run live experiments but instead run offline simulations on existing data to gather valuable insights and compare different promotions. • Business: Businesses do not want to risk live experiments and want to increase profitability.
Preconditions	• The promotions that have been run in the past and their results must be available along with the historical customer data. • There should be a module for offline A/B testing deployed and ready to be utilized.
Success Guarantee (Postcondition)	• The end-user of our system i.e. the marketer, runs simulations on data to attain insights into which promotion is better of the two to maximize profitability and customer satisfaction/engagement.

Main Success Scenario	Actor Action	System Responsibility
	The A/B testing feature is accessed by the end-user(marketer).	
	The marketer selects two or more promotions to compare.	The system simulates the results by analyzing historical data without running live experiments.
	Without running any live experiments, the system uses historical customer data and preferences to predict which promotion strategy will perform better.	The system generates and displays a dashboard showing the predicted performance of each promotion.
	The marketer gathers insights from the simulation results.	The insights will be gathered by the marketer and he/she decides which promotional strategy will perform better on what segment.
Extensions (Alternate Flows)	• E1: The data is missing certain data points to run the simulation: – The system will show a warning message that the data points are insufficient. – The user can still run the simulation but will be warned that the results might not be accurate.	
Special Requirements	• The system should try and mimic real-world customer responses. • The system should be able to handle both large and small datasets and should provide results within 10 minutes.	

Technology and Data Variations	• The simulation's accuracy will be affected by the variations in data e.g. different demographics, physical location, etc..
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2.2.9 Use Case 9

Use Case Name	Marketer Approve or Reject Promotions
Scope	AI Promotion Engine
Level	Sub-function
Primary Actor	Marketer
Stakeholders and Interests	• Marketer: The marketer wants to have a say in what promotions go out and what stays the same so that the business policies regarding profit are not violated. • Business: Businesses want to oversee whether the promotions are aligned with the business goals or not.
Preconditions	• The admin/marketer must be logged into the system with the right privileges. • The promotion calendar must be generated already and displayed to be reviewed by the marketer.
Success Guarantee (Postcondition)	• The promotions that the marketer is suspicious about are flagged and revised and the promotions that are good stay the same.

Main Success Scenario	Actor Action	System Responsibility
	The system utilizes the customer data and purchase patterns of different customers to generate personalized promotions for each segment.	
	The marketer views the generated promotions in the form of a promotional calendar.	The system presents the promotion calendar to the marketer.
	The promotions that have not been viewed yet will have been flagged as not viewed or pending approval.	The system flags any promotions that have not been viewed or approved.
	The marketer then reviews the promotion generated by the promotion engine.	The system allows the marketer to inspect the promotion details.
	The promotions that are rejected are then asked for corrections and flagged, and the approved promotions are rolled out.	Approved promotions are automatically rolled out. Rejected promotions are flagged for corrections.

Extensions (Alternate Flows)	• E1: A promotion is rejected by the marketer: – The system asks for the correction in the promotion by allowing the marketer to edit fields. – The system also logs the correction which can be used to further fine-tune the model in the future.
Special Requirements	• The approval must be easy to navigate.. • The fields of the promotion generated must be editable. • The logs should maintain an audit trail of the changes made.
Technology and Data Variations	• Notifications of pending approvals delivered to the marketers via email or in-app messages.

2.2.10 Module 1 - Promotion Creation

1. The system shall consider and analyze the data available and shall generate promotions specifically for each customer segment.
2. Corresponding to the generated promotions, the system shall create a visually appealing and easy to understand promotion calendar which will clearly show the type of promotion and the value of promotion corresponding to each customer segment.
3. The system shall select the discount type and optimize the discount value by taking into account customer preferences, behavior, and purchasing patterns.

2.2.11 Module 2 - Automated Emails and Analytics

1. The system shall be able to send promotional emails automatically to the right customers based on the generated approved promotional calendar.
2. The system shall provide an insightful dashboard about each rolled out promotion, which the marketer may use to make informed decisions.

2.2.12 Module 3 - System Integration and Deployment

1. The system shall provide RESTful API's for easy integration with numerous and varied e-commerce platforms.
2. The system shall provide a flexible engine, meaning it should be able to handle inputs(data) of different formats.
3. The system shall be scalable, meaning it should be able to handle inputs(data) of different sizes/volumes.

2.2.13 Module 4 - Testing and Optimization

1. The system shall have the option for the user to do offline A/B testing so that the user can run simulations on the existing data before rolling out promotions.
2. The system shall make sure that all blocks/functions work as expected before deployment.
3. The system shall allocate room for feedback-driven fine-tuning of the AI models deployed to maximize and optimize the effectiveness of promotions.

2.3 Non-Functional Requirements

Non-functional requirements highlight the overall quality and robustness of a system. The NFRs for our system (AI-powered promotions engine) are listed below:

2.3.1 Reliability

Reliability is a crucial NFR and it is in place to ensure that the system is robust and if a failure occurs, how the system recovers from it.

1. **NFR1:** The system shall constantly maintain an uptime percentage of 99.5 percent and ensure that the mean time between failures is minimally 5 minutes.
2. **NFR2:** The system shall automatically detect any failures that are about to occur or have occurred and should recover from them with a data loss percentage being at 0 percent.
3. **NFR3:** The system shall maintain logs of all the critical processes that are taking place, and if a failure occurs, the development team has to be notified within 3 minutes.

2.3.2 Usability

For the system to be easy to navigate and intuitive, it has to follow the usability requirements.

1. **NFR4:** The system shall allow the user(marketer) to approve/disapprove any promotion with just a click of a button.
2. **NFR5:** The system shall allow the marketer to change the content of the promotion within 5 minutes.
3. **NFR6:** The system shall display the KPIs in a user-friendly and easy to understand manner so that the marketer understands it in under 30 seconds.
4. **NFR7:** The system shall provide guides on how to use its feature so that a new user can be proficient in using the system within 1 hour of training.

2.3.3 Performance

Performance requirements are there to ensure that the system has high performance and it serves as a measure of how efficiently the system is operating.

1. **NFR8:** The system shall process and store the promotion generated by the engine in the database within 5 seconds of generation.
2. **NFR9:** The system shall store the changes made to the promotions by the marketers in the database within 2 seconds of the "save" button being clicked.
3. **NFR10:** The email promotion push system shall be able to push 50,000 emails within 10 minutes.
4. **NFR11:** The customer segmentation module shall be able to process up to 10 million records and return the segmentation results in under 8 minutes.

2.3.4 Security

For the security of customer data in the system and the prevention of unauthorized access, security requirements have to be followed.

1. **NFR12:** To ensure that the data is not intercepted during transit or it is not compromised during at rest i.e. in the database, it shall be encrypted using standard encryption procedure such as AES-256.

2. **NFR13:** The system shall use multi-factor authentication to authenticate and authorize marketer when accessing sensitive customer/promotion information.
3. **NFR14:** The system shall maintain logs for everyone using it for the past 1 year.

2.3.5 MACH Principles

To focus on the modularity, APIs, cloud scalability, and headless architecture, the system should follow the MACH principles.

1. **NFR15 - Modularity:** The system shall be divided into modules so that new features can be easily added, removed, and/or updated without compromising the working status of the entire system.
2. **NFR16 - API First:** The system shall make use of well-documented RESTful APIs so that multiple platforms can access the promotion engine's functionalities/features seamlessly.
3. **NFR17 - Cloud Native:** The system shall be deployed on the cloud, making it easy to scale up or down and maintain 99.5 percent uptime.
4. **NFR18 - Headless:** The systems backend implementation shall be decoupled so that it can be easily compatible with multiple interfaces e.g. mobile, web, etc.

2.3.6 Scalability

The scalability of our system will ensure that the system can handle input data of various sizes and formats.

1. **NFR19:** The system shall be able to handle input data of Upton 1 million records/entries without reporting degrading in its performance.

2.3.7 Maintainability

It ensures that the system is easy to fix and easy to update for the development team.

1. **NFR20:** The system shall be designed in such a way that new modules can be added and/or updates to existing modules can be made without the need for a complete restart of the system.
2. **NFR21:** The code for the system shall follow proper procedures for comments so that other developers can contribute to it easily with an onboarding time of 3 days.

2.4 Domain Model

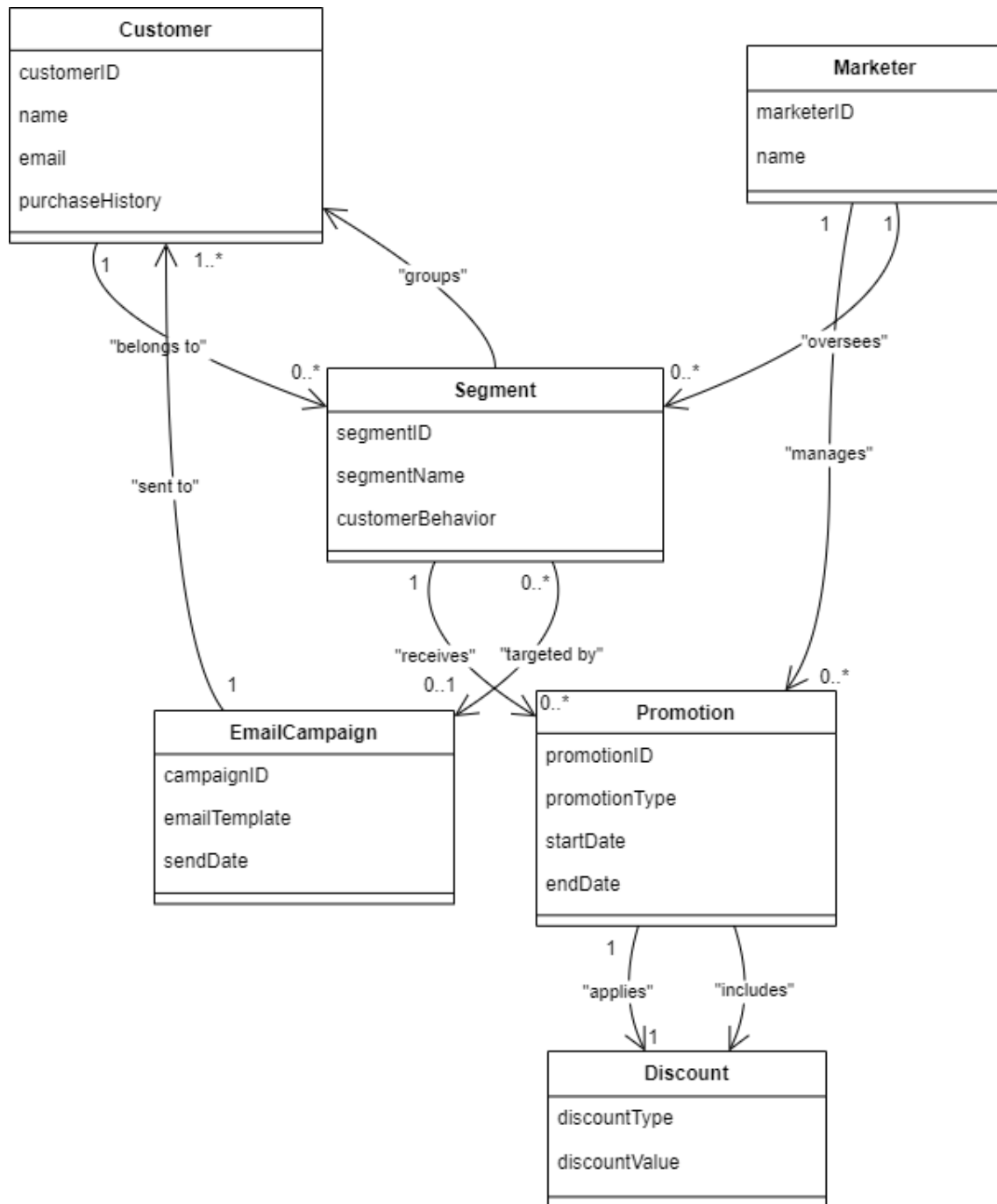


Figure 2.2: Domain Model.

Chapter 3

System Overview

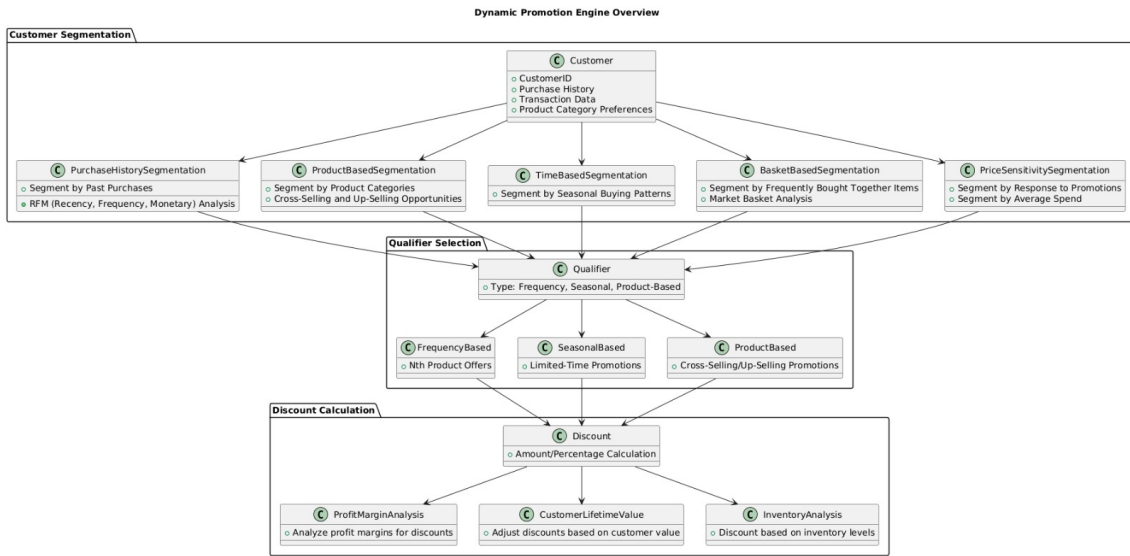


Figure 3.1: General Overview of our system.

3.0.1 Description of System Overview

Our project titled **PromoHive** is an AI powered promotion engine and intends to work on principles of dynamic promotions. Customers have unique preferences and purchasing patterns that are hidden in the huge amounts of data; our engine works by analyzing that customer data. It works by segmenting customers of similar looking purchase history, product category preferences, seasonal trends, cycles and price sensitivity into different clusters or groups. By successful segmentation, we aim to introduce personalization into the promotions the customers receive.

The project works by passing the data into an AI powered promotion engine which consists of 3 main sub-models: customer segmentation, discount type/advanced qualifier cal-

culatation (classification) and discount value/amount calculation (regression/optimization).

This is how the flow looks like: the customers are segmented, then based on each segment's data, the discount type/qualifier is selected e.g. frequency-based, seasonal-based, or product-based, and then finally, considering the profit margins, customer lifetime value, and inventory levels, the right discount value is calculated that ensures that the balance between customer satisfaction and business profitability is maintained.

All in all, PromoHive is designed in such a way that multiple e-commerce companies can use it to boost sales and enhance customer satisfaction by using data to create smart, effective and resonating personalized discounts and promotions.

3.1 Architectural Design

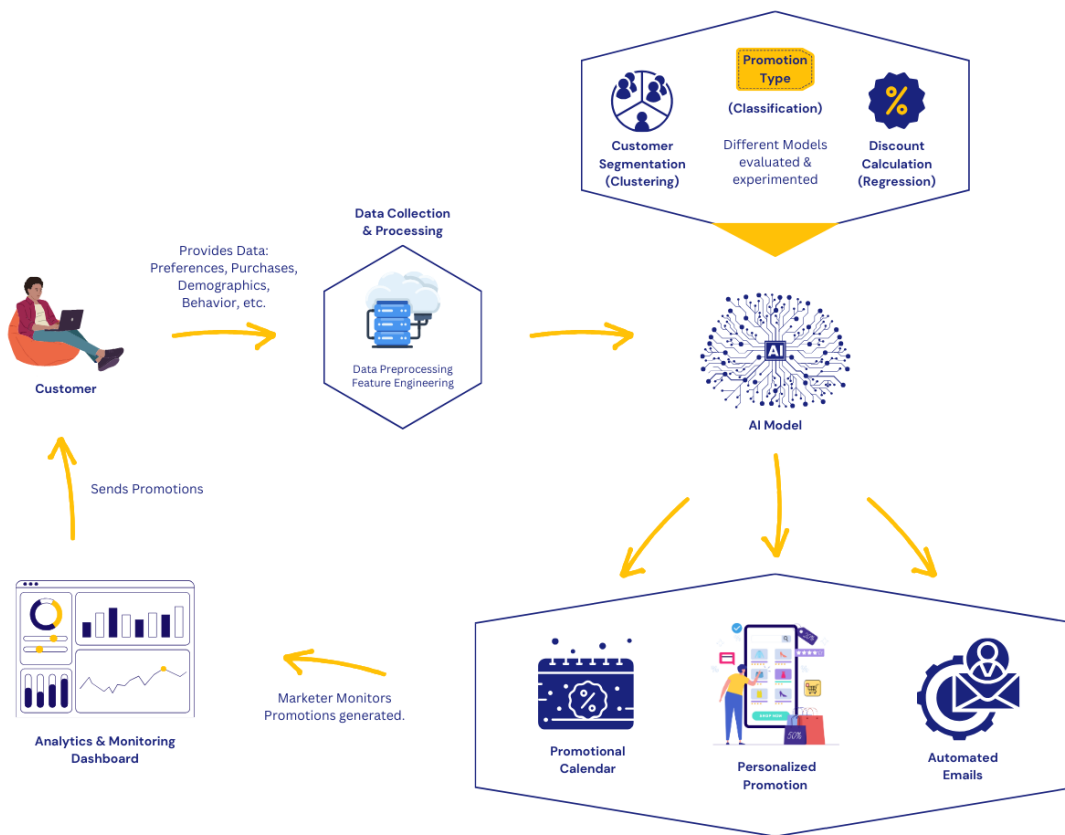


Figure 3.2: Architecture Diagram.

3.1.1 Description of System's Architecture Diagram:

Here is the detailed description of the architecture diagram shown above.

1. **Customer Interaction:** The interaction data of the customer is recorded and the AI model is trained using the help of it. The customer interacts with the website and the data recorded indirectly shows the individual customer's preferences, customer behaviour, purchase patterns, purchase history and demographics which will be used by the AI engine for analysis.
2. **Data Collection and Pre-processing:** In this step, the system will make use of the data collected in previous step and perform data analysis on it along with feature engineering to extract hidden information in the data and to perform analysis.
3. **AI Model:** After recording customer interaction and performing pre-processing on the data, the AI model will perform 3 major tasks:

- **Customer Segmentation:** This part of the AI model will try and cluster customers into groups based on their purchasing behaviour and demographics amongst other characteristics. For this segmentation, different techniques such as K-means clustering and/or DBSCAN will be used.
 - **Promotion Type/Advance Qualifier Selection:** This part of the AI engine will determine what type of discount (BOGO/tiered discount etc.) will be right to target which customer segment using classification techniques.
 - **Discount Value Calculation:** This part of the AI promotion engine will determine the optimum amount of discount to be offered with the selected discount type.
4. **Personalized Promotions:** After the analysis done by the promotional engine, the customers will receive personalized promotions.
 5. **Promotional Calendar:** After the analysis done by the promotional engine, the promotions will be arranged in a calendar clearly showing the time-frame and specific customer segment to target the promotion on.
 6. **Automated Emails:** For the Generated promotions, the system will push out automated emails so that the promotions have been reached to the right customers at the right times.
 7. **Monitoring and Feedback:** The system will track the performance of the promotions and display it in a dashboard for the marketer to see and make informed decisions based on it.

3.2 Design Models

3.2.1 Use Case 1: Generate Promotional Calendar

Activity Diagram:

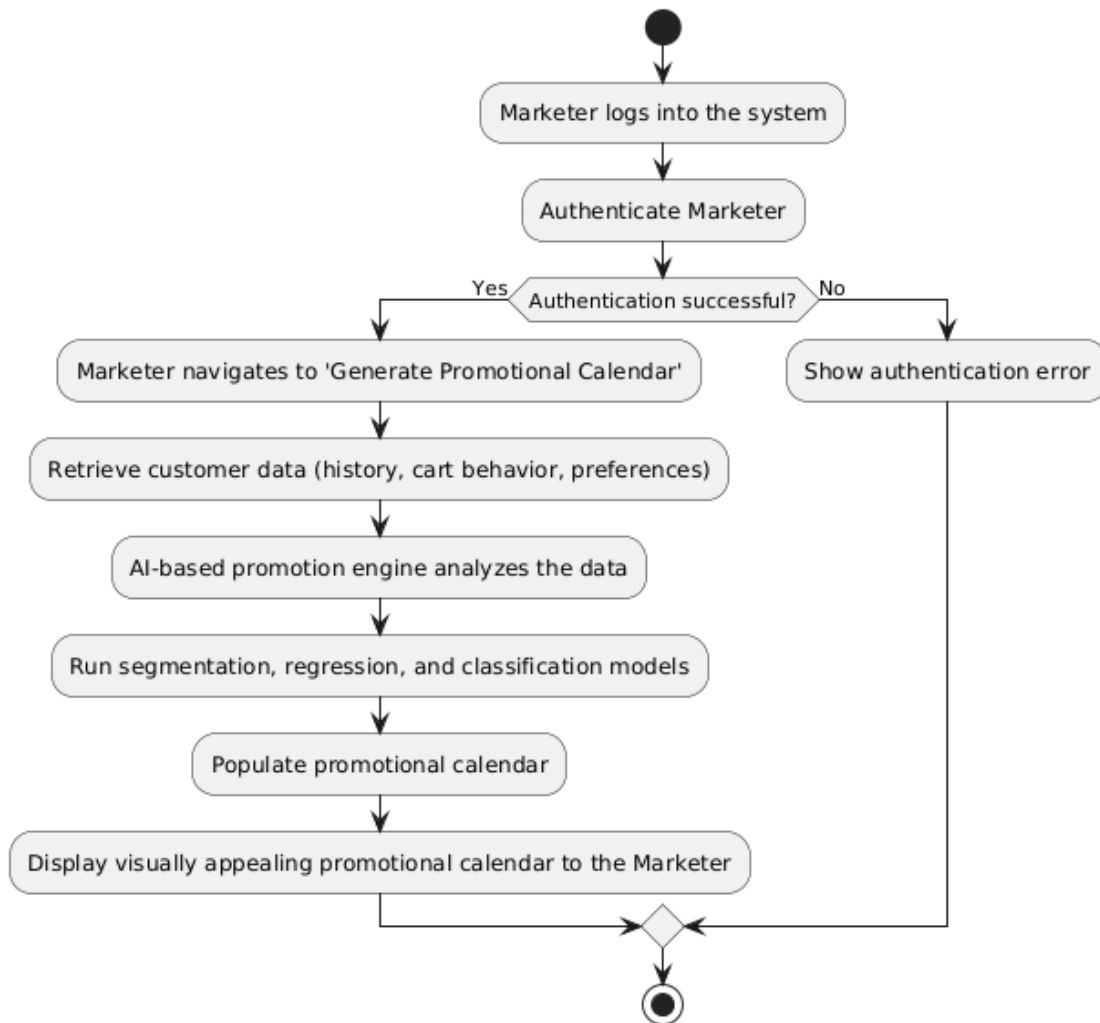


Figure 3.3: Activity Diagram for Use Case 1.

System Sequence Diagram:

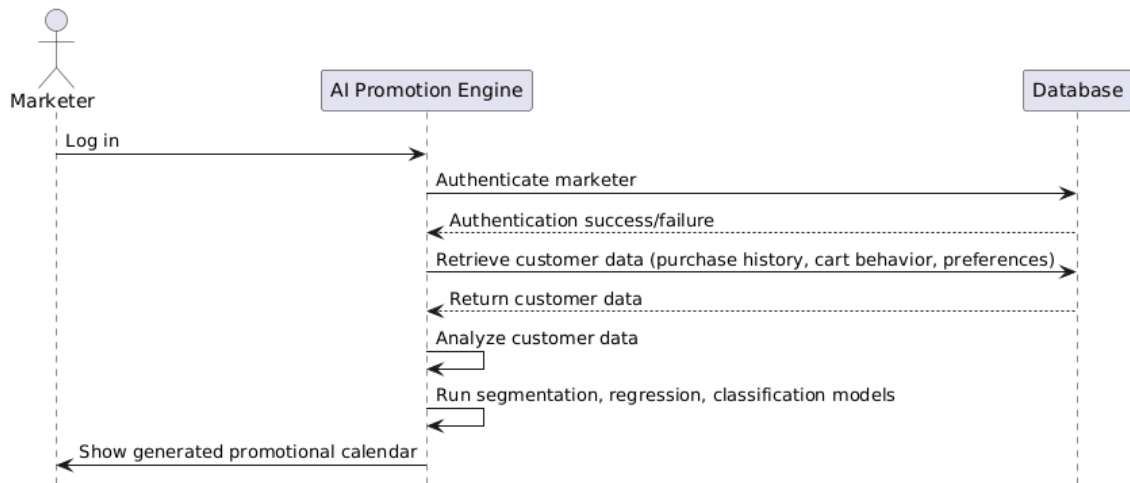


Figure 3.4: SSD for Use Case 1.

3.2.2 Use Case 2: Customer Segmentation

Activity Diagram:

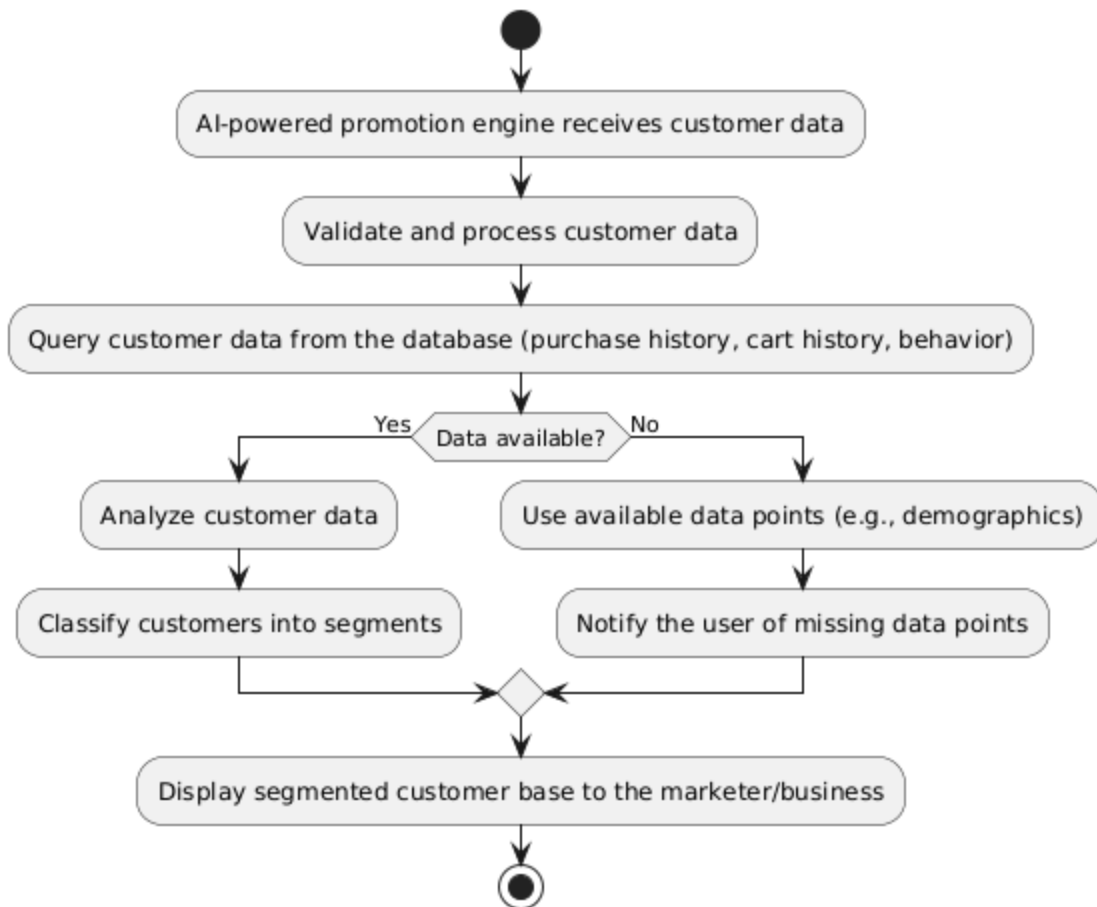


Figure 3.5: Activity Diagram for Use Case 2.

System Sequence Diagram:

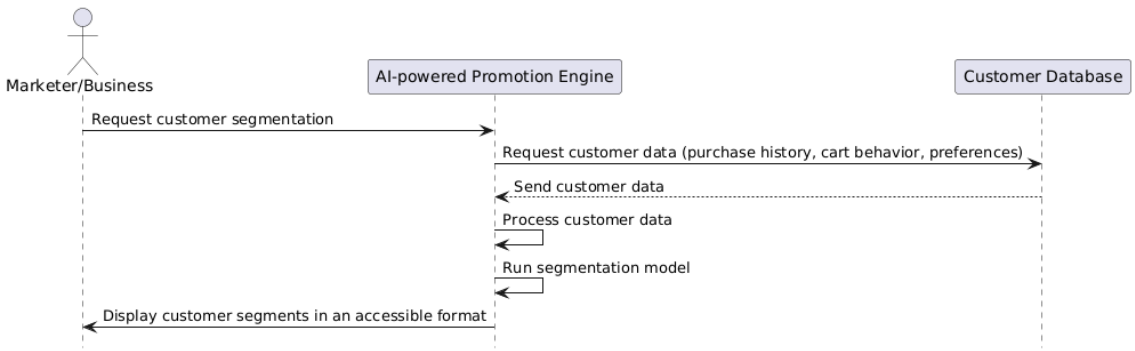


Figure 3.6: SSD for Use Case 2.

3.2.3 Use Case 3: Dynamic Discount Type Classification

Activity Diagram:

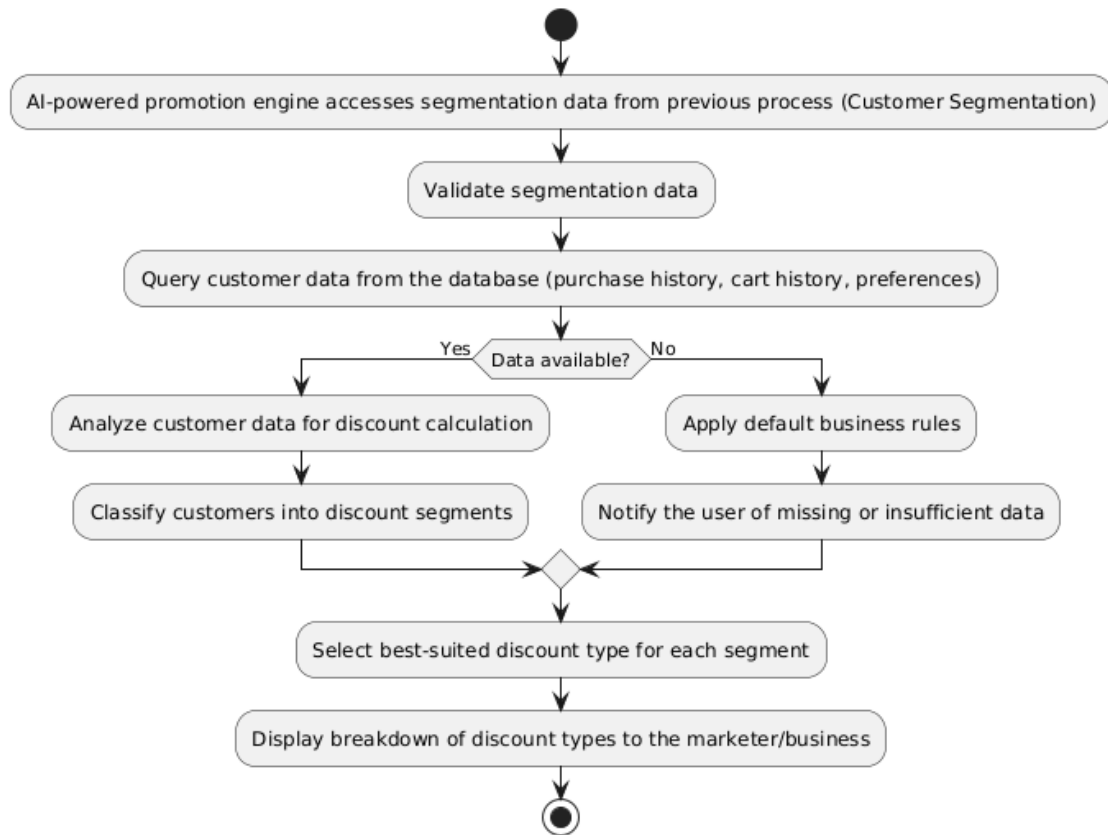


Figure 3.7: Activity Diagram for Use Case 3.

System Sequence Diagram:

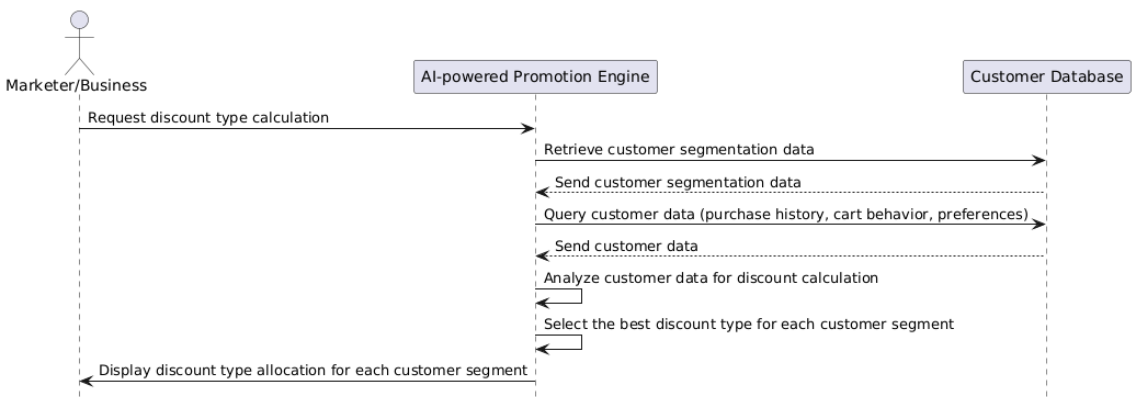


Figure 3.8: SSD for Use Case 3.

3.2.4 Use Case 4: Dynamic Discount Value Calculation

Activity Diagram:

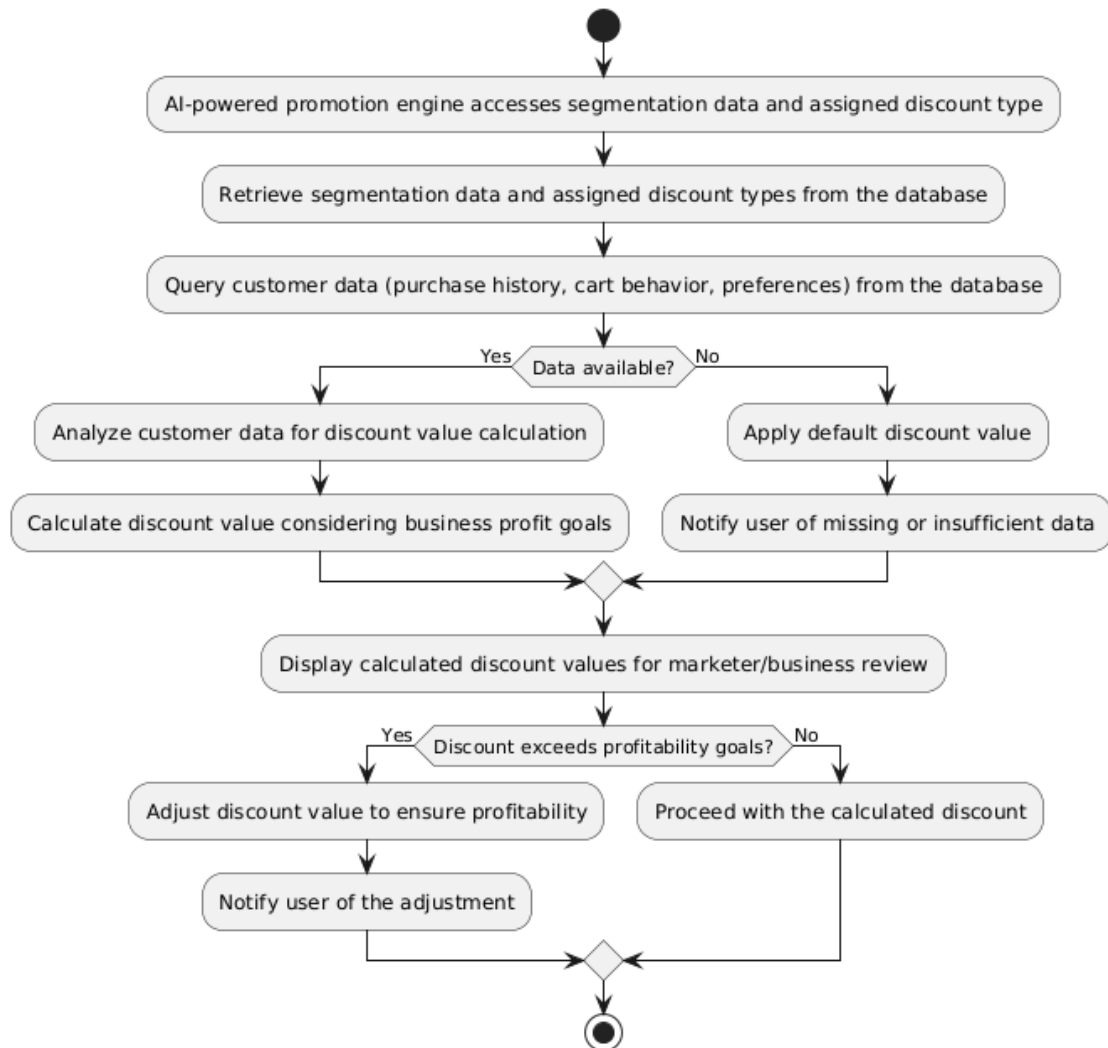


Figure 3.9: Activity Diagram for Use Case 4.

System Sequence Diagram:

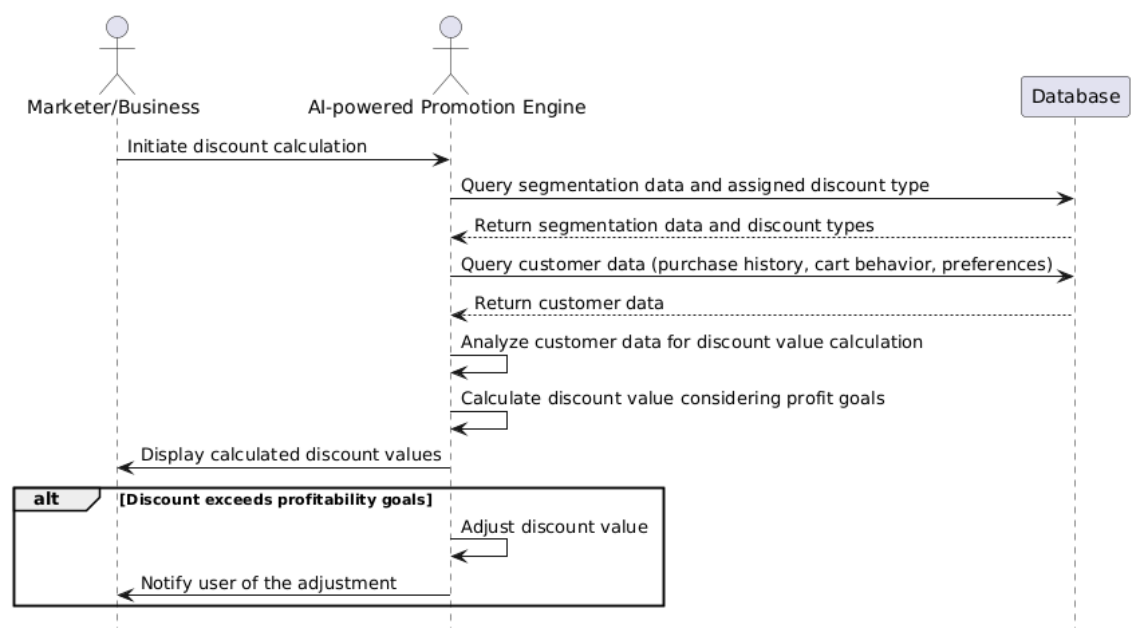


Figure 3.10: SSD for Use Case 4.

3.2.5 Use Case 5: Automated Promotion Execution

Activity Diagram:

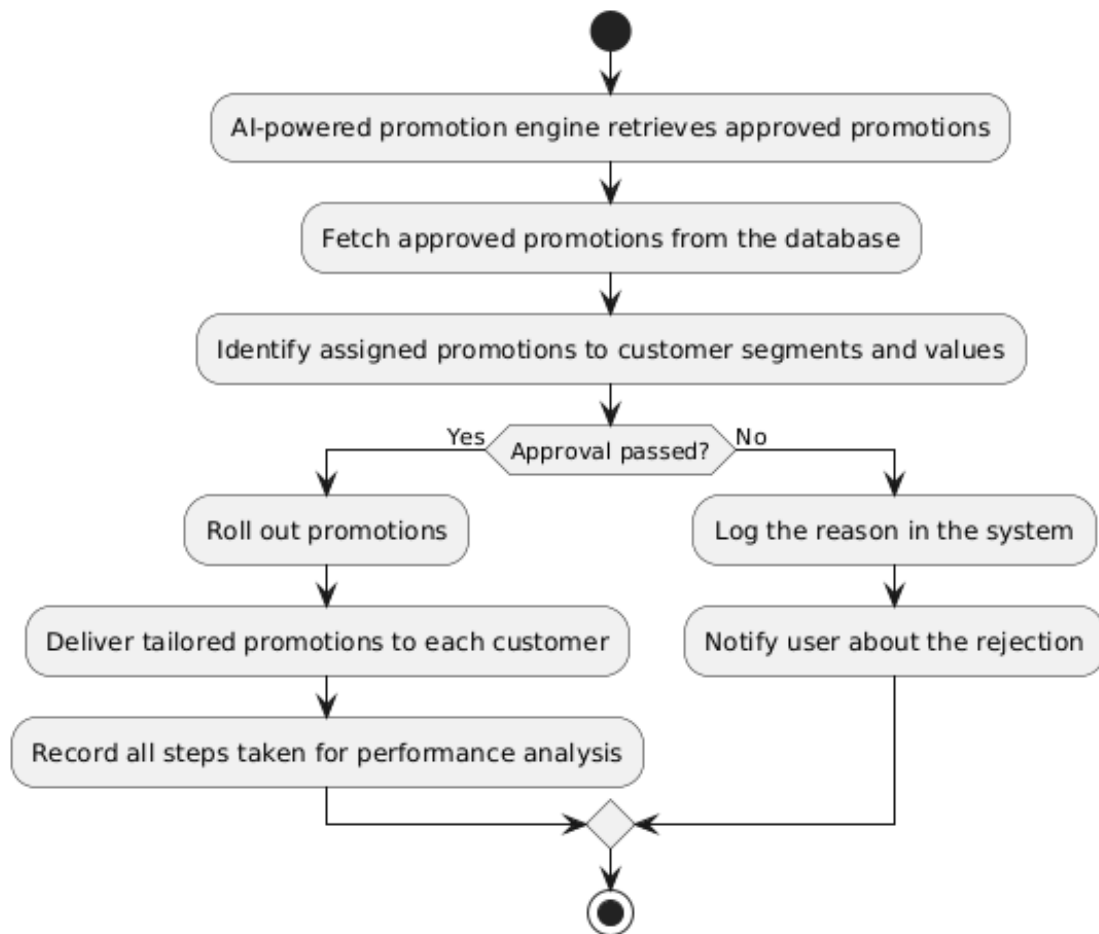


Figure 3.11: Activity Diagram for Use Case 5.

System Sequence Diagram:

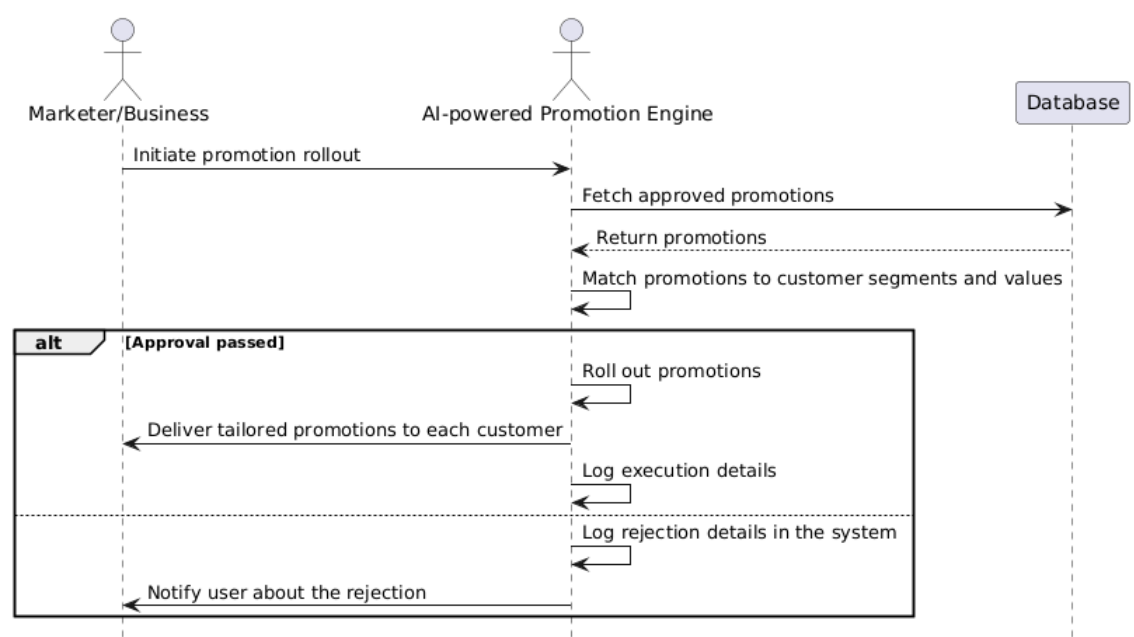


Figure 3.12: SSD for Use Case 5.

3.2.6 Use Case 6: Email Campaigns

Activity Diagram:

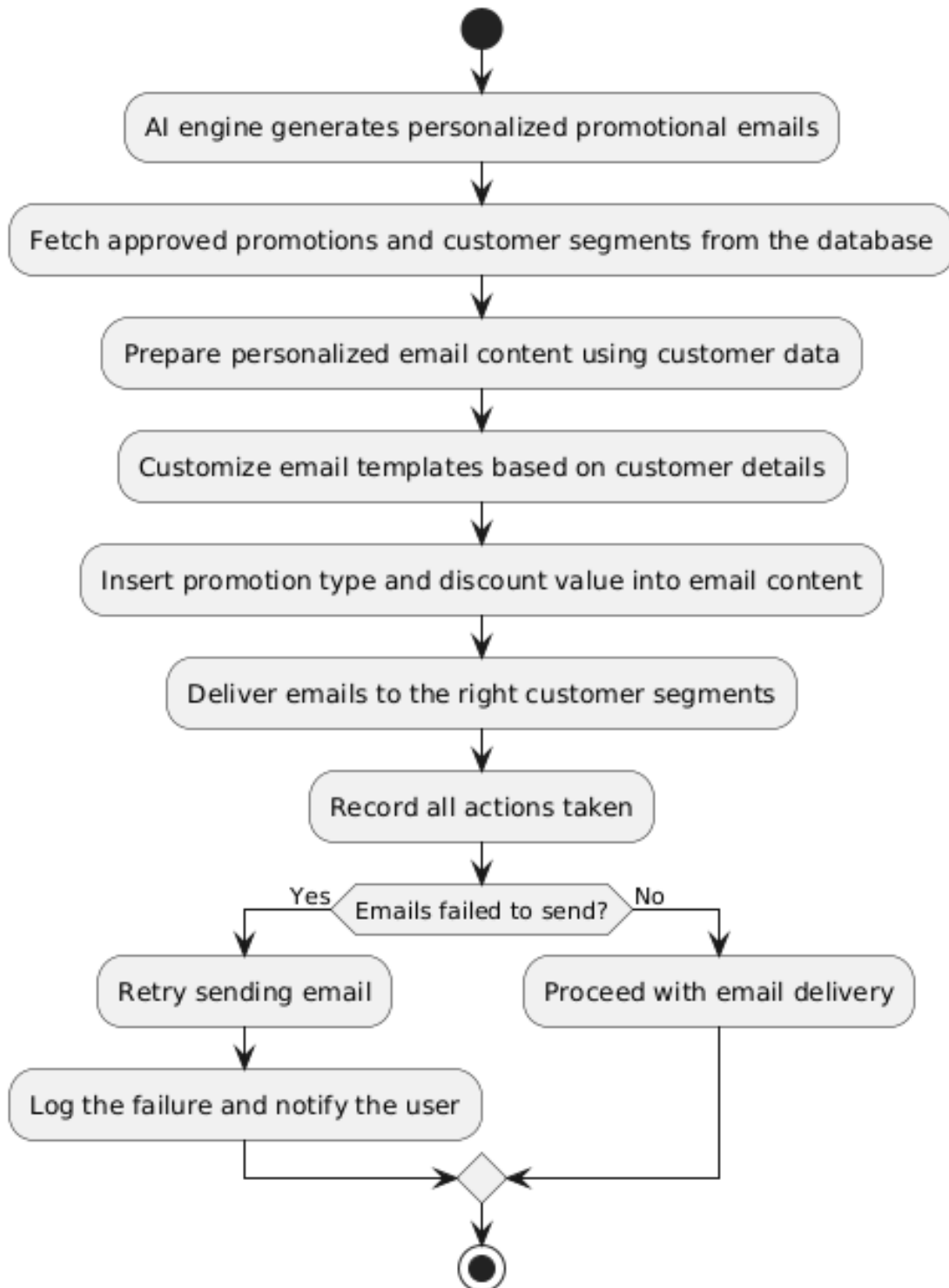


Figure 3.13: Activity Diagram for Use Case 6.

System Sequence Diagram:

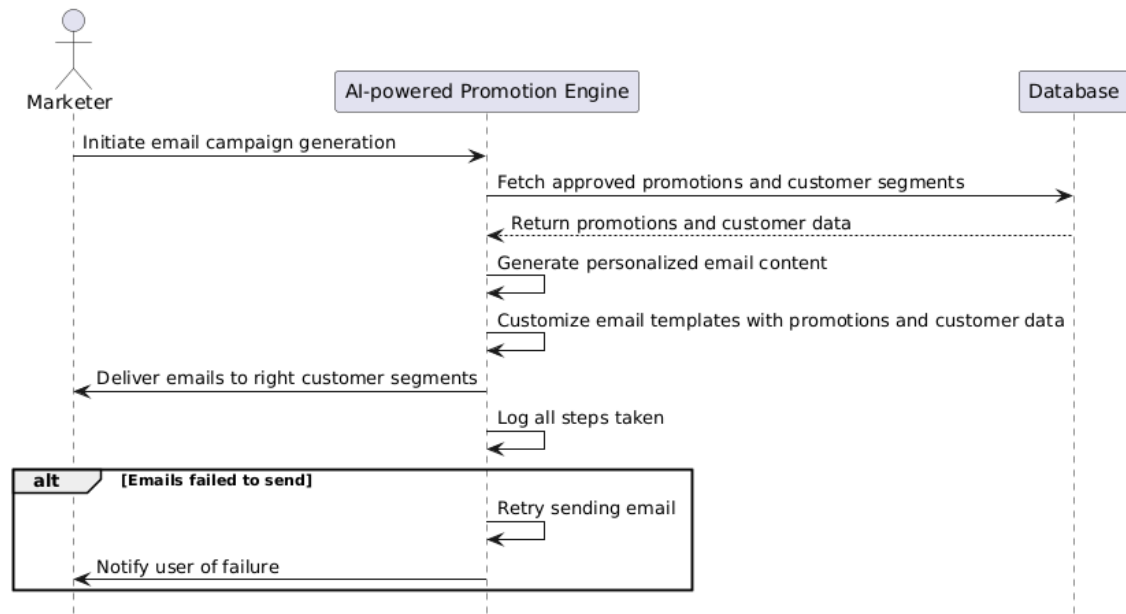


Figure 3.14: SSD for Use Case 6.

3.2.7 Use Case 7: Monitor Promotion Performance

Activity Diagram:

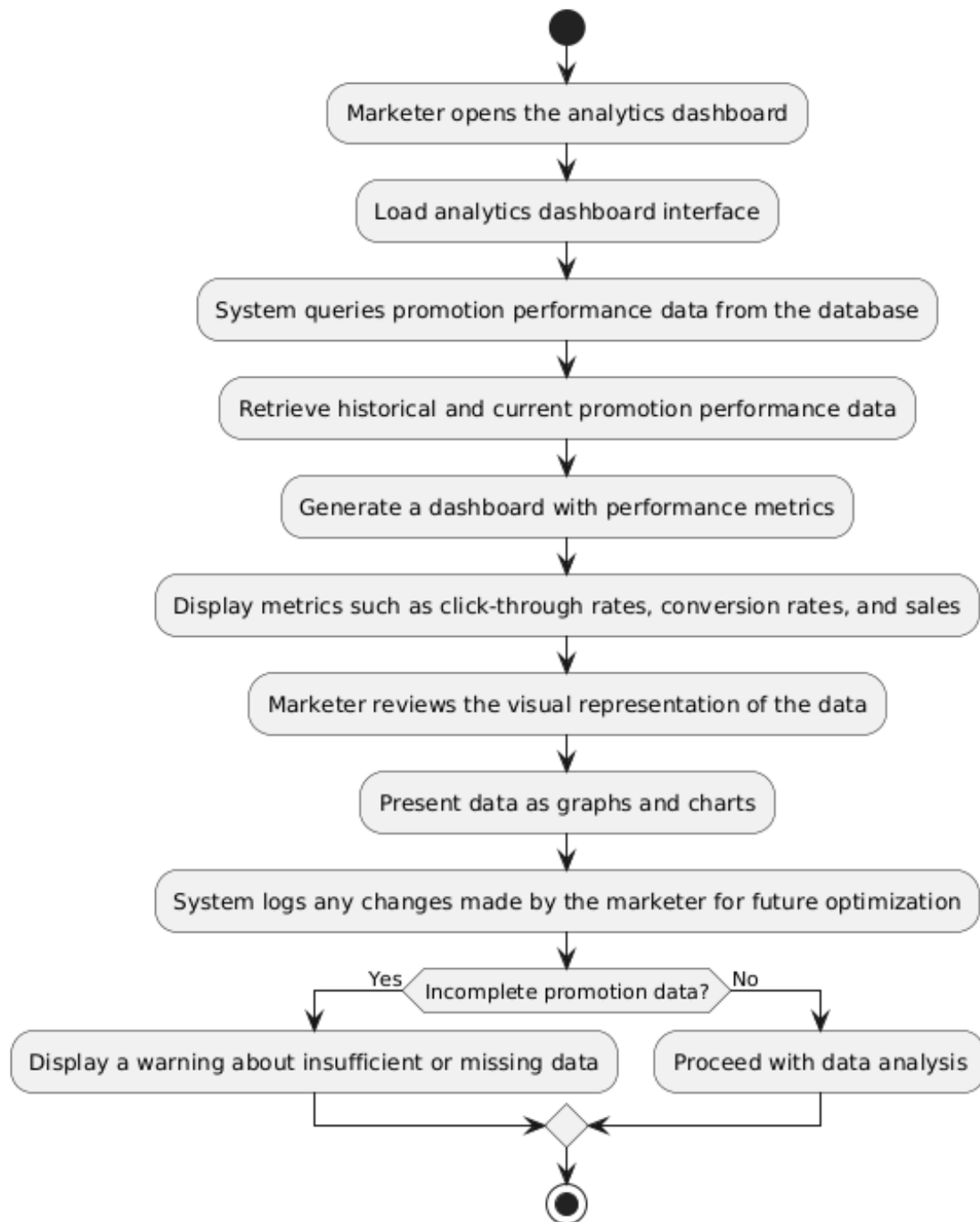


Figure 3.15: Activity Diagram for Use Case 7.

System Sequence Diagram:

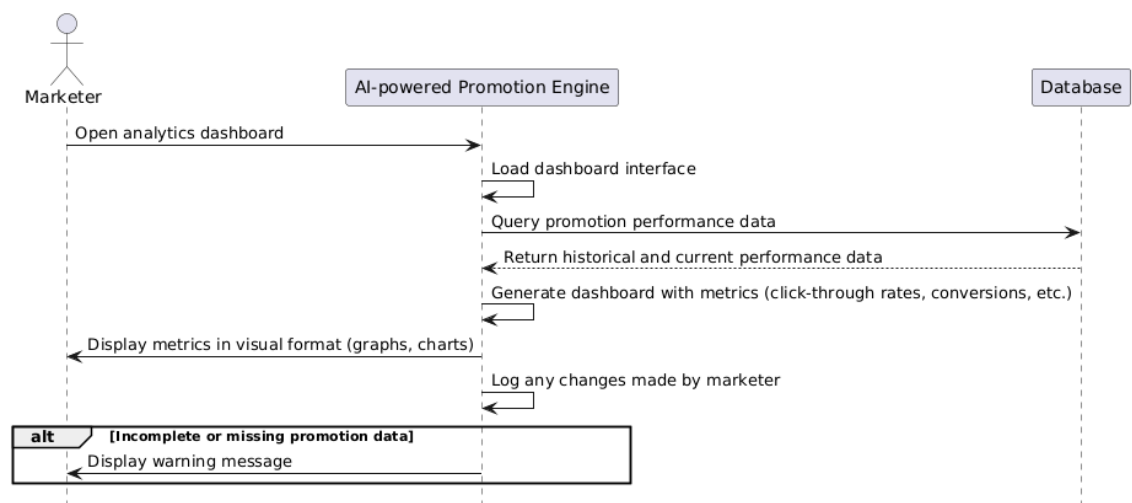


Figure 3.16: SSD for Use Case 7.

3.2.8 Use Case 8: Offline A/B Testing

Activity Diagram:

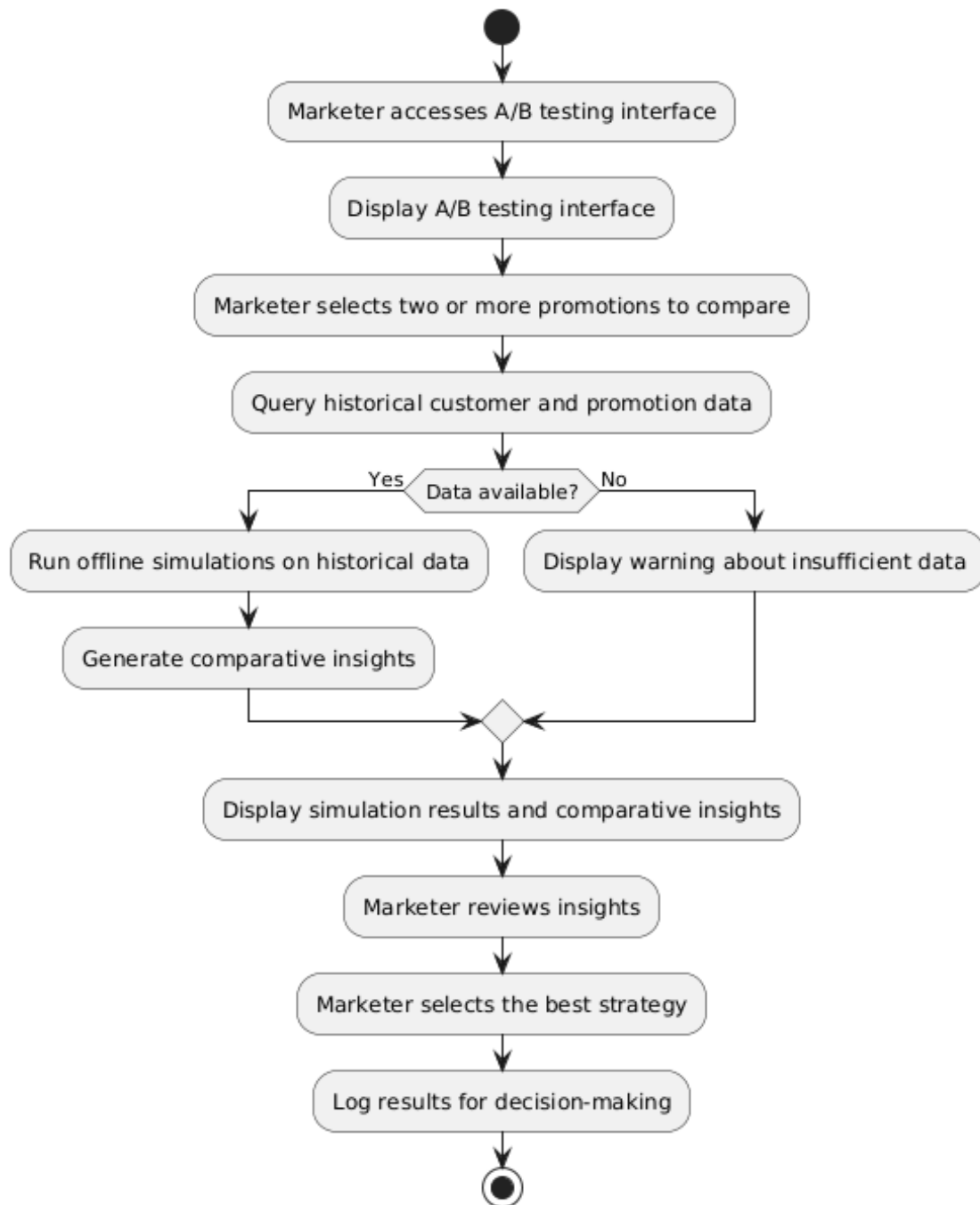


Figure 3.17: Activity Diagram for Use Case 8.

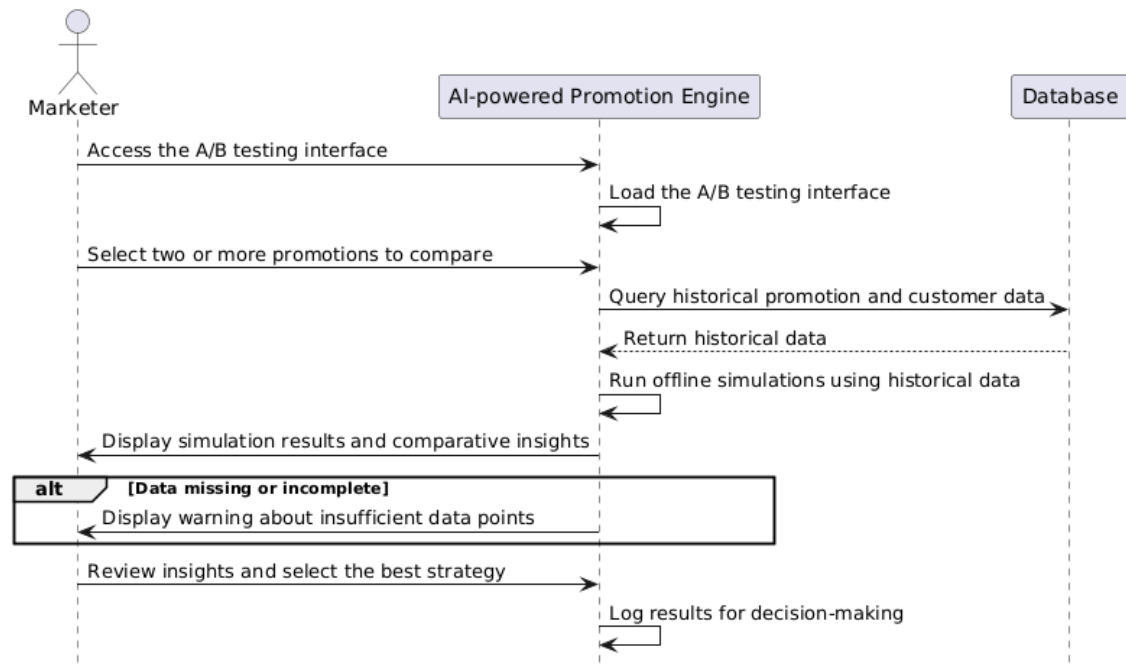
System Sequence Diagram:

Figure 3.18: SSD for Use Case 8.

3.2.9 Use Case 9: Marketer Approve or Reject Promotions

Activity Diagram:

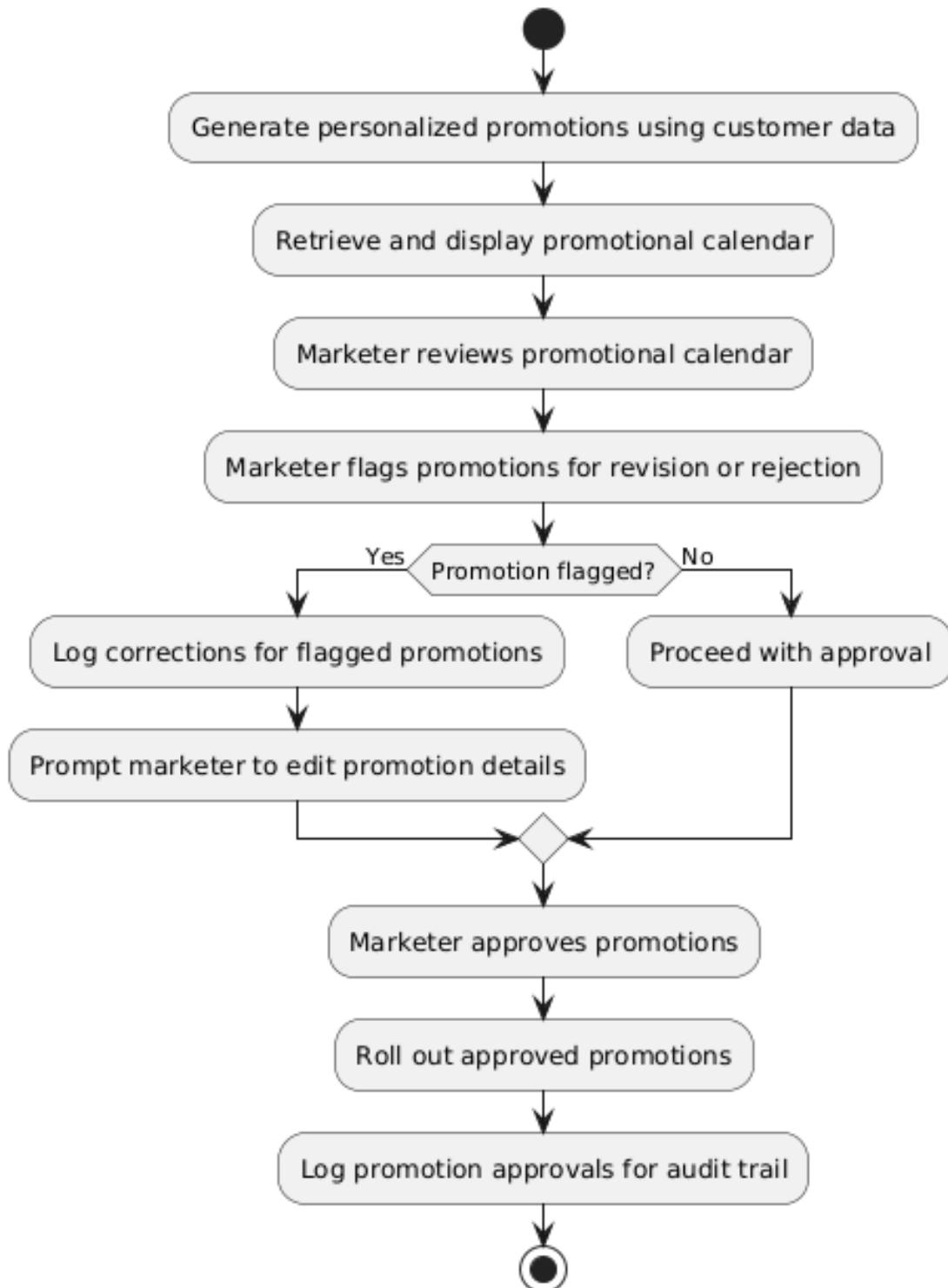


Figure 3.19: Activity Diagram for Use Case 9.

System Sequence Diagram:

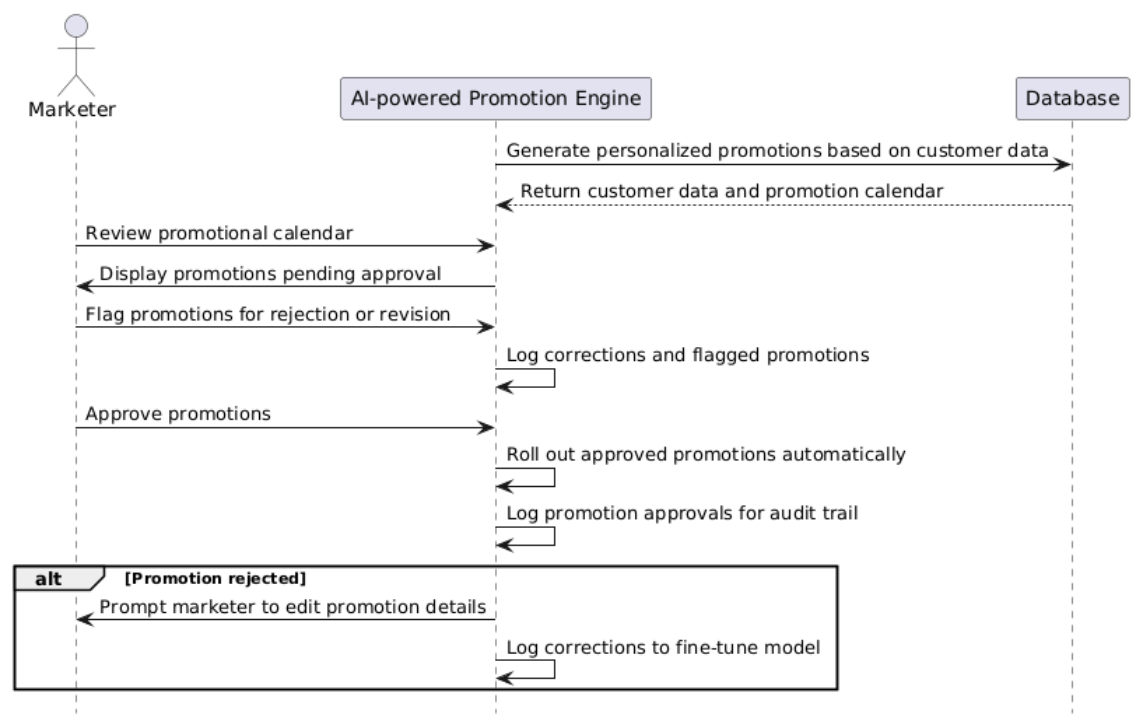


Figure 3.20: SSD for Use Case 9.

3.2.10 System-Level Sequence Diagram:

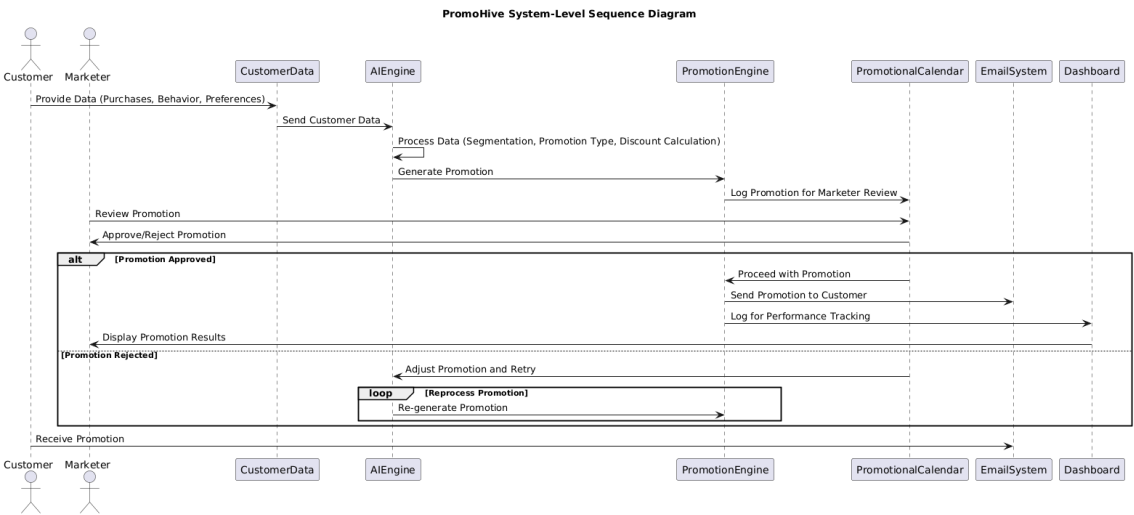


Figure 3.21: System-Level SSD.

3.2.11 Data Flow Diagram:

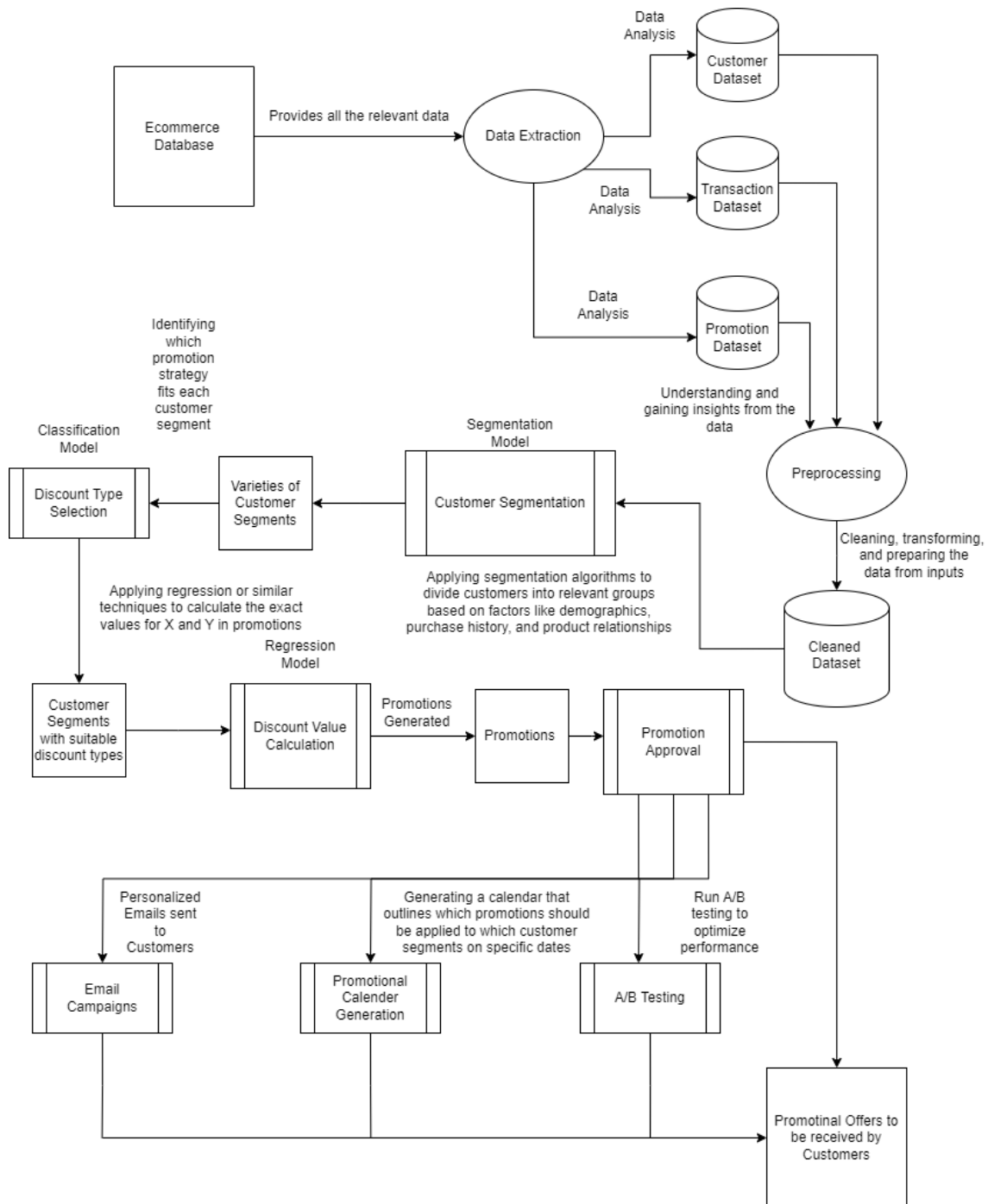


Figure 3.22: Data Flow Diagram of our System.

3.3 Data Design

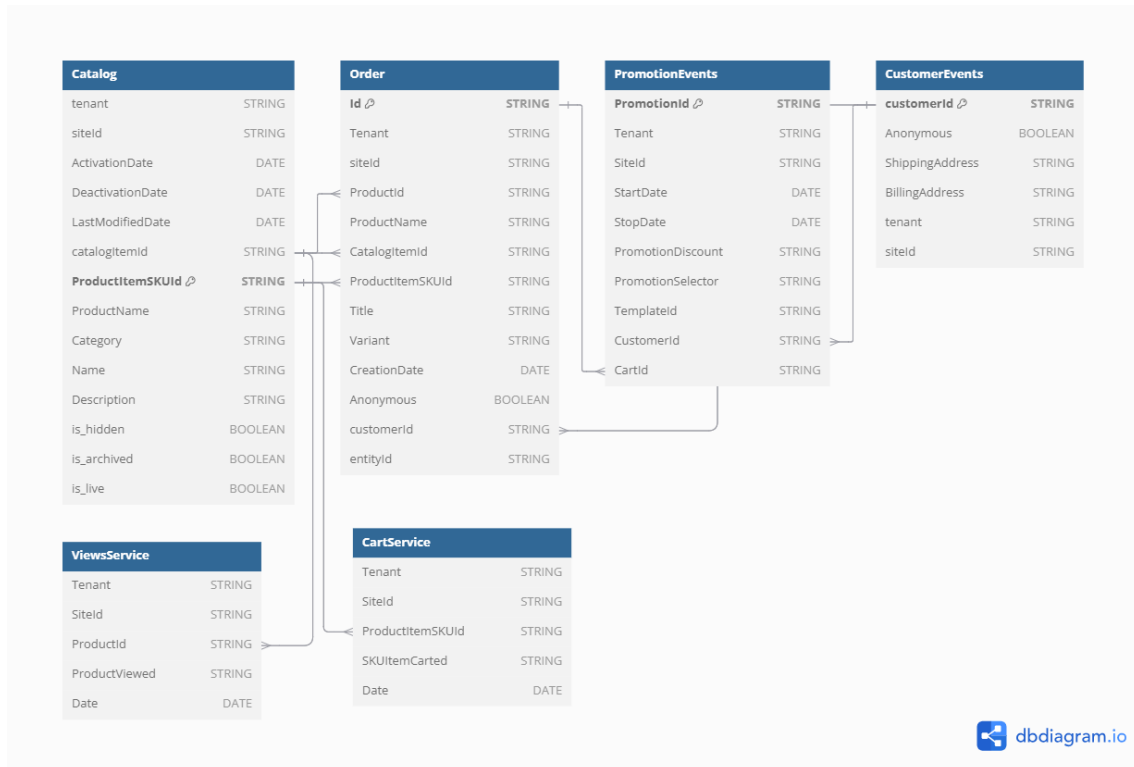


Figure 3.23: System's ERD.

3.3.1 Description of ERD

Below is the description of Entity Relationship Diagram of our system which will handle data from various e-commerce platforms.

1. **Catalog:** This table in the data is responsible for handling the inventory aspect. It includes values like name, category, and description, along with the product's status (live, hidden, archived). It also includes a unique product identifier known as "ProductItemSKUId" which will be used to link product sold with its description.
2. **Order Data:** This is a very important table in the data and it is responsible for recording the purchases made by each customer along with its details such as date and amount.
3. **Promotions Events:** This table is significant in the sense that it stores the information of all the promotions that have been made live in the past along with the number of times each promotion has been availed. It can be linked to the customer using PromotionID.

4. **Customer Events:** This is a table which has crucial information of the customer. It has information of billing address, email and whether the customer has chosen to stay anonymous or not.
5. **Views Service:** This table is a result of a logger, which logs in information of the customer related to product as it stores product details of the product viewed by the customer and the date of the view.
6. **Cart Service:** This table contains data points related to the cart of a customer, it has critical information such as ProductItemSKUId, the cart action, and the date.

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